JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 97, NO. B13, PAGES 19,617–19,639, DECEMBER 10, 1992

Paleomagnetism of the Late Triassic Hound Island Volcanics: Revisited

PETER HAEUSSLER[1] AND ROBERT S. COE

Earth Sciences Board and Institute of Tectonics, University of California, Santa Cruz

T. C. ONSTOTT

Department of Geology and Geophysics, Princeton University, Princeton, New Jersey

The collision and accretion of the Alexander terrane profoundly influenced the geologic history of Alaska and western Canada; however, the terrane's displacement history is only poorly constrained by sparse paleomagnetic studies. We studied the paleomagnetism of the Hound Island Volcanics in order to evaluate the location of the Alexander terrane in Late Triassic time. We collected 618 samples at 102 sites in and near the Keku Strait, Alaska, from the Late Triassic Hound Island Volcanics, the Permian Pybus Formation, and 23-Ma gabbroic intrusions. We found three components of magnetization in the Hound Island Volcanics. The high-temperature component (component A) resides in hematite and magnetite and was found only in highly oxidized lava flows in a geographically restricted area. We think it is primary, or acquired soon after eruption of the lavas, principally because the directions pass a fold test. The paleolatitude indicated by this component (19.2° ± 10.3°) is similar to those determined for various portions of Wrangellia, consistent with the geologic interpretation that the Alexander terrane was with the Wrangellia terrane in Late Triassic time. We found two overprint directions in the Hound Island Volcanics. Component B was acquired 23 m.y. ago due to intrusion of gabbroic dikes and sills. This interpretation is indicated by the similarity of upper-hemisphere directions in the Hound Island Volcanics to those in the gabbro. Component C, found in both the Hound Island Volcanics and the Permian Pybus Formation, is oriented northeast and down, fails a regional fold test, and was acquired after regional deformation around 90 to 100 Ma. This overprint direction yields a paleolatitude similar to, but slightly higher than, slightly older rocks from the Coast Plutonic Complex, suggesting that the Alexander terrane was displaced 17° in early Late Cretaceous time. The occurrence of these two separate overprinting events provides a satisfying explanation of the earlier puzzling results from the Hound Island Volcanics (Hillhouse and Grommé, 1980). Finally, great-circle analysis of the paleomagnetic data from the Pybus Formation suggests the Alexander terrane may have been in the northern hemisphere in Permian time.

INTRODUCTION

Paleomagnetic data from volcanic rocks have demonstrated the far-traveled nature of tectonostratigraphic terranes that make up much of the North American Cordillera. For example, the studies of *Hillhouse* [1977], *Yole and Irving* [1980], *Hillhouse et al.* [1982], and *Hillhouse and Grommé* [1984] indicate paleolatitudes of 11° to 20° and northward displacements of up to at least 2500 km relative to the North American craton for the various fragments of the Wrangellia terrane since Late Triassic time (Table 1). These findings are also consistent with geological interpretations [*Jones et al.*, 1977].

However, in a paleomagnetic study of the Late Triassic Hound Island Volcanics of the Alexander terrane, Wrangellia's large neighbor to the southeast (Figure 1), *Hillhouse and Grommé* [1980] found no indication of poleward displacement relative to North America, implying that the Alexander terrane lay far north of Wrangellia during the Late Triassic. This result was troublesome because geologic mapping and geochemical data presented by *Gardner et al.* [1988] suggest a plutonic tie between the two terranes in Pennsylvanian time, and paleontological evidence suggested that they occupied similar paleolatitudes in the Late Triassic [*Newton*, 1983]. An improved Late Triassic reference pole for North America put forward by *May and Butler* [1986] places the craton farther to the south than was believed earlier but of course does not change the discrepancy in paleolatitudes for Wrangellia and the Alexander terrane derived from the paleomagnetic studies.

Despite the severe paleogeographic inconsistencies imposed by the tectonic interpretation of the paleomagnetic results from the Hound Island Volcanics, the data appeared very good [*Hillhouse and Grommé*, 1980]. The natural remanent magnetization (NRM) of the samples exhibited fairly high coercivities and unblocking temperatures during progressive alternating field (AF) and thermal demagnetization, and it decayed univectorially toward the origin after removal of a less stable component that was similar in direction to the present-day field. These characteristic (e.g., highest temperature or coercivity) directions clustered well within flows, yielding precision parameters typical of primary remanent magnetization found in young lavas. Finally, the flow means were of both polarities and passed a fold test.

In order to investigate the discrepancy between paleomagnetic results for the Wrangellia and Alexander terranes, we resampled the Hound Island Volcanics for a new paleomagnetic study. We were able to sample over a larger region than was done in the original study, we sampled other lithologies, and we carried out $^{40}Ar/^{39}Ar$ analyses on some of the samples we collected. In this paper we shall show how the additional information we obtained has enabled us to find a satisfying resolution of the paradox. For those readers who are familiar with the tectonic framework of the Canadian and Alaskan Cordillera and do not wish to be concerned with the details of the paleomagnetic story, we encourage you to skip ahead to the section entitled "Tectonic Implications.'

TECTONIC AND GEOLOGIC SETTING

The collision and accretion of allochthonous terranes dominates the Mesozoic and Cenozoic geologic history of Alaska. The most significant

[1]Now at U.S. Geological Survey, Anchorage, Alaska.

Copyright 1992 by the American Geophysical Union.

Paper number 92JB01361.
0148-0227/92/92JB-01361$05.00

TABLE 1. Late Triassic Paleolatitudes With Inferred Tectonic Displacements and Rotations For Wrangellia and the Alexander Terrane Assuming Location in the Northern Hemisphere

Locality	Observed Paleolatitude	Expected Paleolatitude (North)	Displacement, km	Rotation (Clockwise)
Wrangell Mountains, Alaska [*Hillhouse*, 1977]	10.5° ± 4.6°	33.5° ± 5.5°	2553	99.2° ± 15.1°
Alaska Range, Alaska [*Hillhouse and Gromme*, 1984]	13.9° ± 3.8°	35.6° ± 5.5°	2409	89°-166°
Vancouver Island, British Columbia [*Yole and Irving*, 1980]	18.0° ± 4.7°	19.6° ± 5.5°	177	-158.3° ± 10.8°
"Seven Devils," Southeast, Oregon [*Hillhouse et al.*, 1982]	19.3° ± 13.3°	13.6° ± 5.5°	-633	-57.8° ± 20.7°
Keku Strait, AK (this study)	19.2° ± 10.3°	29.3° ± 5.5°	1121	138.0° ± 12.6°

terrane to dock with southern Alaska and western Canada is the amalgamated terrane sometimes referred to as "Baja British Columbia" [*Irving et al.*, 1985]. Although this name has genetic implications that cannot be demonstrated by existing paleomagnetic studies, we will use it in this paper. This superterrane consists of two amalgamated terranes (Figure 1): (1) the Insular Belt, composed of the Peninsular, Wrangellia, and Alexander terranes, and (2) the Intermontane Belt, composed of the Stikine, Quesnel, Cache Creek, and several smaller terranes [*Monger et al.*, 1982].

The Alexander terrane is the largest in the southeastern Alaska. It occupies a central structural position within Baja British Columbia, and the suture between the Insular and Intermontane belts lies along its eastern border. The Alexander terrane has a remarkably long and continuous marine stratigraphic section spanning the entire Paleozoic [*Gehrels and Saleeby*, 1987*a*, *b*]. The overlying Triassic strata, which consist of pillow basalts, basalt flows, breccia, rhyolites, and limestones, are interpreted by *Gehrels and Saleeby* [1987*b*] as by-products of a rifting event. Jura-Cretaceous flysch and volcaniclastic and volcanic rocks succeed the Triassic strata. This group of rocks, which was deformed and shortened in early Late Cretaceous time, is called the Gravina-Nutzotin Belt [*Berg et al.*, 1972]. It lies on the eastern margin of the Alexander terrane and laps onto Wrangellia (Figure 1).

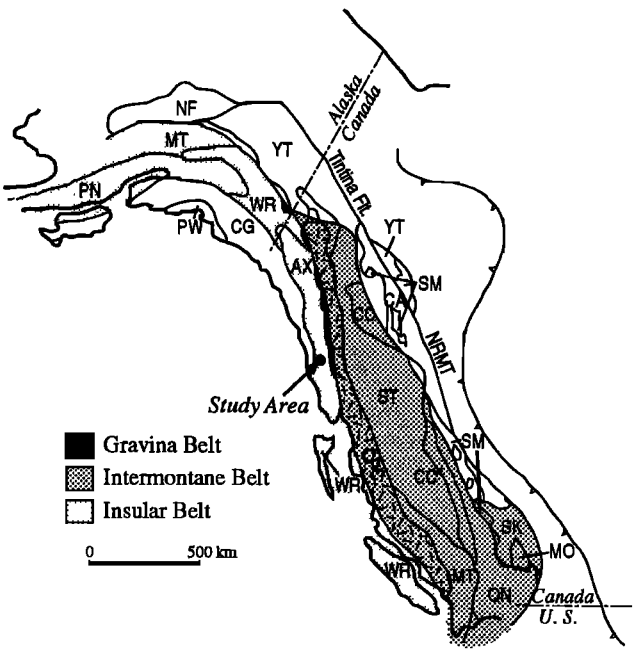


Fig. 1. Major terranes of western Canada and southern Alaska. Terrane abbreviations are AX, Alexander; CA, Cassiar; CC, Cache Creek; CG, Chugach; MT, Miniterranes; MO, Monashee; NF, Nixon Fork; PN, Peninsular; PW, Prince William; QN, Quesnellia; SK, Selkirk; SM, Slide Mountain; ST, Stikine; WR, Wrangellia; YT, Yukon-Tanana; SM, Slide Mountain. NRMT, Northern Rocky Mountain Trench. Coast Plutonic Complex (CPC) is designated by the random hatchure pattern. The figure is modified from *Umhoefer* [1987].

Wrangellia lies immediately north of the Alexander terrane within the Insular Belt (Figure 1) [*Berg et al.*, 1972; *Jones et al.*, 1977; *Nockleberg et al.*, 1985]. Wrangellia was apparently broken up and dispersed along the western North American continental margin, with the result that slivers of the terrane are found from the Wrangell Mountains in southern Alaska to Vancouver Island, British Columbia, and possibly as far south as Hells Canyon in Idaho [*Jones et al.*, 1977]. Wrangellia and the Alexander terrane must have been amalgamated before Late Jurassic time, because the Gravina-Nutzotin belt overlaps both terranes. Moreover, *Gardner et al.* [1988] suggested a substantially earlier tie, because a Pennsylvanian syenitic complex apparently intruded both terranes.

The Peninsular terrane is the third terrane in the Insular Belt and lies west of Wrangellia (Figure 1). These two terranes were probably in close proximity by Late Jurassic time, because Jura-Cretaceous flysch is found along the northern border of both terranes [*Jones and Silberling*, 1979]. *Detterman and Reed* [1980] argue that the Peninsular and Wrangellian Triassic stratigraphies are correlative, indicating an earlier tie between the terranes, whereas *Jones and Silberling* [1979] believe that the differences between the pre-Late Jurassic strata of the terranes are greater than their similarities.

In summary, the terranes of the Insular Belt were amalgamated at least by Late Jurassic time and may have been together throughout much more of their geologic history.

The Intermontane Belt consists of the Stikine, Cache Creek, and Quesnellia terranes and other miniterranes (Figure 1). The Stikine and Quesnellia terranes have volcanic arc affinities, whereas the Cache Creek terrane and other, smaller miniterranes have oceanic affinities. Sedimentary overlap assemblages indicate these terranes were amalgamated by the end of the Triassic. In Late Jurassic time this superterrane collided with North America and formed the Omineca metamorphic and plutonic Crystalline Belt along its eastern margin [*Monger et al.*, 1982].

During Late Jurassic-Early Cretaceous time, the terranes in the Insular Belt probably formed a long, skinny landmass with two trough like flysch basins on the northern and eastern sides. Likewise, the

Intermontane Belt terranes probably formed a long landmass. During Late Cretaceous time, the Insular Belt apparently collided with the previously accreted Intermontane Belt, as indicated by the multitude of Late Cretaceous and early Tertiary plutons of the Coast (Ranges) Plutonic Complex that intrude both terranes (see Figure 1) [*Monger et al.*, 1982].

The Hound Island Volcanics

The Hound Island Volcanics are the most abundant member of the Late Triassic Hyd Group, which crops out in the Keku Strait in southeastern Alaska (Figure 2) [*Muffler*, 1967]. The Hound Island Volcanics consist of interbedded basaltic pillow breccia and pillow lava, lava flows, andesitic pyroclastic breccia, and minor limestone beds [*Muffler*, 1967]. The limestone interbeds provide age-diagnostic macrofossils and commonly contain abundant *Halobia* and/or *Monotis* fragments, all of which indicate a late Carnian to late Norian age. All evidence indicates shallow submarine deposition of the Hound Island Volcanics, because the lava flows are interbedded with pillow lavas and some siltstones and fossiliferous limestone.

The major deformation in the Keku Strait probably occurred between ~100 and 90 Ma for the following reasons. The Jura-Cretaceous Gravina-Nutzotin belt rocks in the Keku Strait are folded, and 90-Ma granitoids near Petersburg, Alaska, also intrude folded rocks of the Gravina belt [*Brew and Morrell*, 1983]; thus the deformation is older than 90 Ma. The onset of deformation was certainly by Albian time, because there are common folded Albian age rocks in the Gravina belt [*Berg et al.*, 1972]. However, recent reports of Cenomainian age Gravina belt rocks [D. Brew, oral communication, 1990; *Cohen et al.*, 1990] suggest the deformation began a few million years later, or the deformation may have occurred synchronously with deposition of the youngest Gravina sedimentary rocks. In any case, the major deformational pulse probably occurred after Albian time. According to the *Harland et al.* [1982] time scale, the Albian-Aptian boundary is at 98 Ma. It is reasonable to assume that the deformation that folded the Gravina belt rocks also folded the rocks in the Keku Strait [*Muffler*, 1967].

McClelland and Gehrels [1990] have recently documented an Early to Middle Jurassic deformation to the east of the Keku Strait. This deformation is apparently related to right-lateral shear along the eastern boundary of the Alexander terrane and probably did not affect the rocks in the Keku Strait.

The Hound Island Volcanics have been metamorphosed to subgreenschist facies. Optical and electron microscopy show these rocks contain a primary igneous texture. Many rocks show albitization and sericitic alteration of the plagioclase (grains are ≤ 10 μm). Most rocks have unidentified green minerals (an Fe, Mg aluminosilicate (nontronite?)) as an alteration around the margins or completely replacing igneous clinopyroxene or olivine. We also observe prehnite, chlorite, sphene, ferric biotite (1x10 μm), quartz, calcite, and iddingsite. We did not observe epidote. Despite the presence of these metamorphic minerals, *Brew and Muffler* [1965] report the presence of undevitrified volcanic glass in the Hound Island Volcanics on Hound Island (Figure 2). However, a K/Ar date of 126 Ma from carefully selected shards of the glass suggests amorphous alteration.

Although the major regional deformation occurred in Late Cretaceous time, we think that the metamorphism of the Hound Island Volcanics occurred on the seafloor soon after eruption. Our main reasons are that chlorite, calcite, prehnite, and other unidentified green minerals fill vesicles. The $^{40}Ar/^{39}Ar$ analyses on samples of the Hound Island Volcanics, which we discuss later in this paper, also indicate that the major metamorphism occurred soon after deposition. Finally, an anomalously large $^{87}Sr/^{86}Sr$ ratio from one sample on Hound Island (Figure 2) is most easily understood if there was intensive seawater interaction with the lava [*Samson et al.*, 1989], consistent with hydrothermal alteration of the lavas on the seafloor.

The Point Camden Gabbro

The Hound Island Volcanics were intruded by 23-Ma gabbroic intrusions in the Keku Strait. We will informally refer to this unit as the Point Camden gabbro because of outcrop exposures at Point Camden on Kuiu Island (Figure 2). We have studied the paleomagnetism of these intrusions [*Haeussler et al.* this issue]; however, we evaluate the effects of these intrusions on the paleomagnetism of the Hound Island Volcanics in this paper.

PALEOMAGNETIC SAMPLING METHODS

We sampled the Hound Island Volcanics over the entire area where they are exposed, and we sampled other units we thought would be helpful in understanding the area's paleomagnetic and thermal history (Figure 2). We collected samples from 70 sites in the Hound Island Volcanics, eight sites in the Pybus Formation (dolomite), and 24 sites in the Point Camden gabbro. Our standard practice was to collect six samples at every locality; however, we collected 10 or 11 samples when drilling clasts in pillow breccias. In total, we obtained 618 cores at 102 sites. We determined the orientations of samples by using a magnetic compass and inclinometer, checking the compass orientations with a solar compass when possible. The bedding attitude of the lava flows was planar on the scale of the outcrops and was easily measured. Sections of pillow lavas were never thick enough to preclude obtaining bedding attitudes from lava flows or sediments above or below the section that was sampled. The bedding attitude we use for each site is the *Fisher* [1953] mean of five measurements.

We spent portions of two field seasons collecting samples, changing our sampling procedure of the Hound Island Volcanics slightly from the first year to the second. Some of the lava flows at Cape Bendel are thin (≤ 30 cm) and appear to have been erupted in quick succession. The first summer we drilled only one sample per flow for each group of flows at a given locality (sites 1-19). Our results indicate there generally are not large differences in directions among flows at a given locality, except at one site. There is a substantial difference in directions among samples from site 10, and thus the sample directions are given as the flow directions in Table 2. The second summer, we drilled six or more samples from each flow, which we considered one site.

PALEOMAGNETIC RESULTS

Hound Island Volcanics

We performed detailed thermal demagnetization on all samples and alternating field (AF) demagnetization on companion specimens from at least one core in most of the first 20 sites. We used a three-axis Schonstedt tumbling AF demagnetizer and a Schonstedt oven (all specimens heated in air) in a shielded room at the University of California at Santa Cruz. All measurements were made with a 2G three-axis cryogenic magnetometer. Through these initial investigations we found that thermal demagnetization was better at isolating characteristic remanence directions than AF demagnetization. Therefore we used thermal demagnetization on samples from most sites. Nevertheless, heated samples usually acquired a greater isothermal remanent magnetization (IRM) than did unheated samples. This relationship suggests some alteration occurred upon heating, such as the conversion of pyrite or pyrrhotite, which are observed in thin section, to magnetite. Despite this effect, the remanence directions were well constrained, and the advantages of thermal cleaning appeared to outweigh the disadvantages. Because of substantial differences in blocking temperatures between sites any sort of

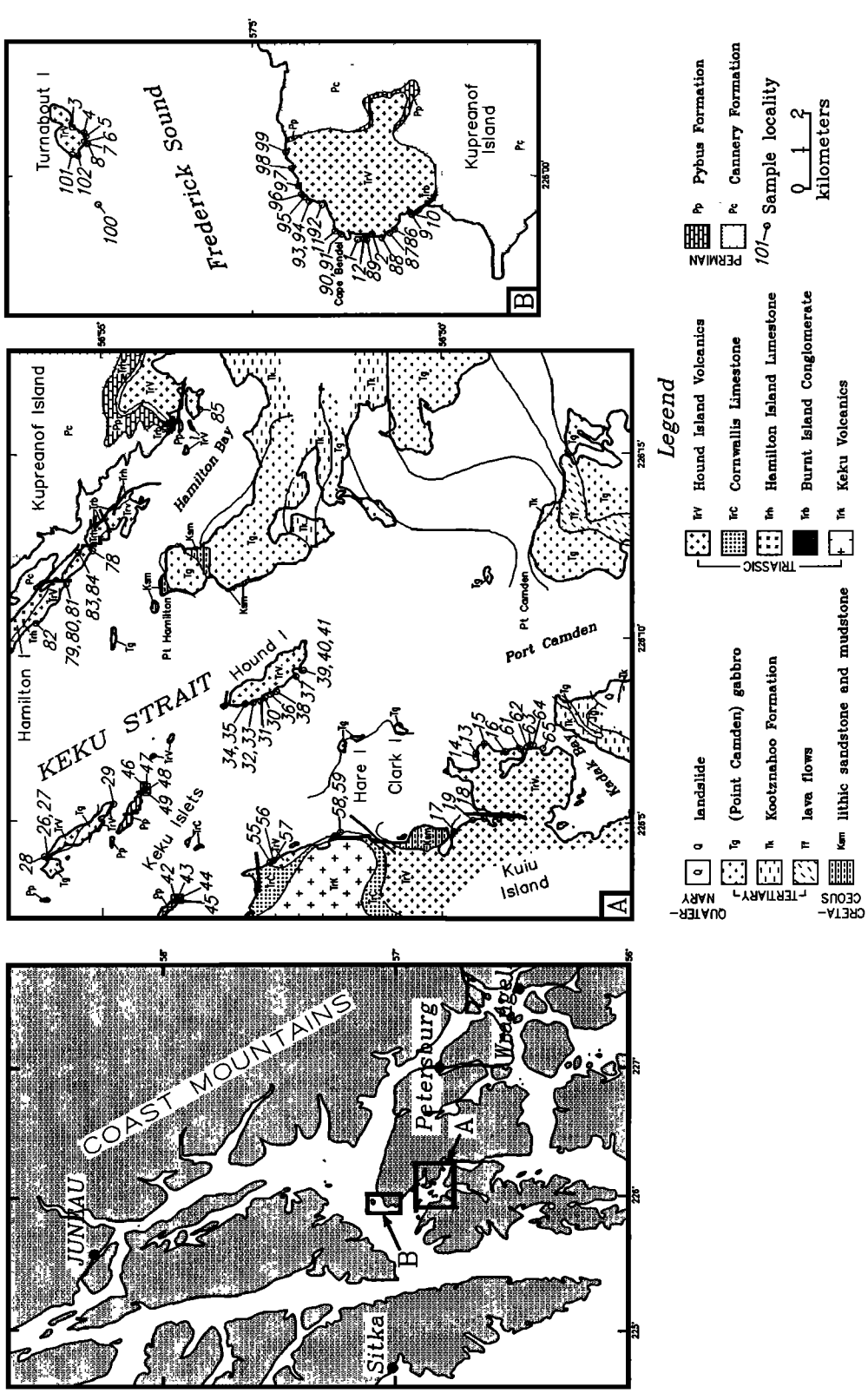


Fig. 2. Location of the sampling area in southeastern Alaska and the paleomagnetic sampling sites in the Keku Strait. Sites in the Hound Island Volcanics and in the Pybus Formation are denoted with circles and squares, respectively. The geologic map is from *Muffler* [1967]. The position of maps A and B are shown on the location map at left, and the scale of both maps is the same.

blanket demagnetization approach was inappropriate. All specimens had at least eight thermal demagnetization steps, and the average number was 13. The earlier study of *Hillhouse and Gromme* [1980] relied chiefly upon AF demagnetization. In the cases where we sampled the same sites as *Hillhouse and Gromme* [1980], we obtained significantly different directions from about half of the sites using thermal demagnetization.

After demagnetization of the specimens, we plotted vector diagrams [*Zijderveld*, 1967] for each sample to examine different components of magnetization isolated by progressive demagnetization. We found from one to three components of magnetization, and we determined a least squares fit of the portion of the vector diagram that decayed univectorially for each component in each sample. Often the lowest blocking temperature component (usually ≤ 300°C) was randomly oriented, suggesting it was an isothermal or viscous remanence acquired during handling or storage. Less often, we observed a component that was approximately parallel to the present-day geomagnetic field (PDF) direction at blocking temperatures < 350°C. There is some difficulty in recognizing components formed by remagnetization in the PDF, because the PDF direction is similar to another component isolated at higher temperatures. In all cases where we could distinguish secondary components, they were removed at blocking temperatures ≤ 450°C.

In addition to the low blocking temperature components described above, we found three characteristic directions of magnetization in the Hound Island Volcanics (Tables 2 and 3). For ease in referring to these components we will assign letter names to them. In in situ coordinates, they are (1) a northwest and shallow direction (component A), (2) a southerly and steeply upward direction (component B), and (3) a northeast and steeply downward direction (component C). Figure 3 shows representative Zijderveld diagrams of these three directions.

Component A: Northwest and Shallow Direction

Directional data. We found component A (northwest and shallow direction) in the Hound Island Volcanics exposed at Cape Bendel on Kupreanof Island, on Turnabout Island, and also on one of the Keku Islets (Figure 2). This component was not observed by *Hillhouse and Gromme* [1980], nor did we find it at any of the sites we sampled that were previously studied by *Hillhouse and Gromme* [1980], which are 14 km south of Cape Bendel. In addition, we found this component only in rocks from lava flows and pyroclastic deposits, not in pillow lavas and pillow breccias.

The component A directions pass a fold test, indicating the magnetization was acquired before folding of the strata. The data pass the standard regional tilt test of *McElhinny* [1964] at 82% confidence (Figure 4). There is no area where a simple two-limb fold test can be performed, but the greater clustering of directions on a stereonet after unfolding the strata is clearly visible to the eye. Another more rigorous way of evaluating the significance of the structural correction is to use a newly developed fold test by *McFadden* [1990]. In this test, one examines the correlation between paleomagnetic directions, the direction of their bedding corrections, and the in situ and tilt-corrected mean directions. The correlation is measured as an angle between the dip direction and the vector between the mean and the sample direction. The sum of the cosines (referred to as *SCOS*) of the angles is then computed. When this value exceeds a critical value for 95% confidence, then the paleomagnetic directions are said to be "correlated with the tectonics." Therefore a significant correlation of the paleomagnetic directions with the dip direction before structural correction, but not after, comprises a positive fold test. For the Hound Island Volcanics in situ direction, with $N = 18$, the *SCOS* value is 7.528. The 95% critical value of *SCOS* is 4.940, and the 99% critical value is 6.919. Thus, there is a strong correlation (> 99% confidence) between the dip direction and the in situ direction in the Hound Island Volcanics. There is no significant correlation between the tilt-corrected directions and the tectonics. In addition, the minimum *SCOS* and the maximum k (e.g., lowest correlation with dip direction and maximum clustering of directions) values are at 100% unfolding, indicating that the magnetization is truly prefolding.

TABLE 2. Paleomagnetic Data for Component A in the Hound Island Volcanics

Site	Rock Type	N	Comment	R	α_{95}	k	I	D	Ic	Dc	Strike/Dip
1	Fs	4/6	R	3.97860	7.8	140.2	2.0	321.9	-16.2	323.7	194.7/23.0
2	Fs	3/6	R	2.88809	30.1	17.9	-6.1	335.7	-14.0	334.3	289.5/11.0
3	PC	1/6	N	---	---	---	-2.4	107.9	37.9	104.7	209.6/41.3
4	Fs	6/6	R,1L-5P	5.88458	10.3	43.3	-30.8	264.6	-54.1	257.4	189.4/24.5
5	Fs	6/6	R	5.92968	8.0	71.1	-11.3	274.2	-40.1	270.3	197.5/29.8
6	Fs	3/6	R, 2L-5P	2.97964	12.5	98.2	-9.7	278.6	-48.1	275.8	194.5/38.7
10	F	#55	R	---	---	---	-6.3	315.7	1.7	314.4	329.0/32.5
10	F	#56	R	---	---	---	-31.9	333.7	-28.9	314.1	329.0/32.5
10	F	#57	R	---	---	---	-63.0	349.0	-56.6	289.8	329.0/32.5
10	F	#58	R	---	---	---	-55.6	46.3	-82.8	338.8	329.0/32.5
10	F	#59,60	R	1.99557	16.7	225.8	-38.9	341.8	-38.5	314.8	329.0/32.5
11	Fs	6/6	R	4.95111	8.5	81.8	-5.0	318.2	-27.7	317.2	236.7/23.0
12	Fs	6/6	R	5.97839	4.4	231.3	-1.3	325.4	-21.4	325.1	237.4/16.5
28	F	5/6	N	4.97813	5.7	182.9	4.8	99.5	-14.2	102.9	308.8/39.5
29	F	5/6	N	4.87683	13.6	32.5	7.4	140.3	21.5	120.6	300.0/68.0
12 and 89	Fs	10/12	R	9.97587	2.5	373.0	-1.6	325.5	-18.1	325.4	237.4/16.5
92	PC	4/6	R	3.99261	4.6	405.7	-25.9	280.2	-40.0	284.8	165.8/15.7
95	F	6/6	R,1L,5P	5.99866	1.1	3737.1	-4.5	270.9	-23.8	271.2	164.9/20.1
96	F	6/6	R,2L,4P	5.96006	6.0	125.2	25.7	270.2	11.1	267.2	147.1/17.1
99	F	6/6	E	5.94724	6.9	94.8	26.8	195.3	6.4	211.3	179.1/26.8
Mean		18/19		14.50181	17.5	4.9	-17.2	304.6	-31.0	298.1	

N is the number of stable samples in a locality out of total number collected (at site 10, #55 and following refer to the sample number), I is in situ inclination, D is in situ declination, Ic is tilt-corrected inclination, and Dc is tilt-corrected declination. Rock type shorthand is Fs, lava flows; F, lava flow; and PC, pyroclastic deposit. Comment code is N, normal polarity; R, reversed polarity (assuming northern hemisphere paleoposition); E, eliminated direction, P and L indicates that combined known directions (lines) and great circles (planes) were used for calculating remanence directions. For example: if we used five planes and one line, we denote it as 5P&1L. The R, α_{95}, and k listed are for the in situ directions. The mean direction after tilt correction is Dec = 297.4°, Inc = -30.7°, $k = 6.7$, $\alpha_{95} = 14.5°$. The mean of the virtual geomagnetic poles gives a paleomagnetic pole of Lat = -2.7°, Long = 104.6°, $K = 7.8$, $A_{95} = 13.2°$. The average site latitude and longitude is 57.1°N, 226.9°E.

TABLE 3. Paleomagnetic Data for Components B and C in the Hound Island Volcanics

Site	Rock Type	N	Comment	R	α_{95}	k	I	D	Ic	Dc	Strike/Dip
1	Fs	6/6	C,F	5.97881	4.4	236.0	65.3	49.0	68.2	352.9	194.7/23.0
2	Fs	5/6	C,F	4.91841	11.0	49.0	62.8	53.0	53.2	44.4	289.5/11.0
3	PC	5/6	C,F	4.96944	6.7	130.9	42.1	102.4	76.3	51.9	209.6/41.3
4	Fs	6/6	C,F	5.87094	10.9	38.7	86.5	70.6	5.6	250.2	189.4/24.5
5	Fs	3/6	C,F	2.99202	7.8	250.7	70.9	23.3	56.8	323.9	197.5/29.8
6	Fs	6/6	C,F	5.88923	10.1	45.1	67.4	49.0	58.9	322.4	194.5/38.7
7	Fs	6/6	C,F	5.92771	8.1	69.2	63.3	60.0	67.5	352.9	199.4/28.6
8	P	5/6	C,F	4.99243	3.3	528.2	46.1	84.9	69.2	55.5	199.4/28.6
9	P	6/6	C,F,6GC's	5.99999	0.1	344672.6	72.5	90.4	34.3	72.5	332.8/39.7
11	PC	5/6	C,F	4.97401	6.2	153.9	63.2	54.0	54.4	17.4	236.7/23.0
13	Fs	4/6	C	3.97092	9.1	103.2	57.3	353.2	27.8	346.4	245.9/30.3
14	P	6/6	C	5.94084	7.3	84.5	50.2	8.9	24.0	357.7	244.2/29.7
15	P	5/6	C	4.98298	5.0	235.0	55.3	2.2	35.9	18.7	320.1/24.8
16	P	4/6	B	3.96529	9.9	86.4	-36.4	214.8	-38.4	200.2	201.6/19.0
17	F	6/6	C	5.90850	9.1	54.6	58.4	0.2	48.8	9.5	310.1/11.8
18	Fs	6/6	C	5.96130	5.9	129.2	51.1	5.0	48.0	347.3	187.3/15.2
19	Fs	5/6	C	4.96899	6.8	129.0	50.6	359.4	17.3	7.2	292.0/35.0
26	S	4/6	C	3.98864	5.7	264.2	75.3	0.2	36.2	29.4	311.4/42.0
27	PC	6/6	C	5.96818	5.4	157.1	78.1	27.3	36.4	37.8	311.4/42.0
28	F	5/6	C	4.92828	10.3	55.8	51.9	68.7	15.5	57.4	308.8/39.5
29	F	3/6	C	2.97329	14.3	74.9	37.1	324.5	-4.6	343.3	300.0/68.0
30	P	5/6	U	4.93612	9.7	62.6	22.7	201.4	32.1	193.5	345.8/17.7
31	PB	10/10	C	9.22980	4.5	116.9	73.7	22.9	60.3	27.3	303.0/13.5
32	F	6/6	B	5.96277	5.8	134.3	-77.7	27.8	-88.4	255.4	303.0/13.5
33	F	6/6	B	5.98347	3.9	302.5	-81.4	18.8	-84.4	235.4	303.0/13.5
34	F	6/6	B	5.97847	4.4	232.2	-81.8	121.7	-70.4	192.3	307.3/18.7
35	F	6/6	B	5.60465	19.6	12.6	-80.7	130.7	-68.7	191.0	307.3/18.7
36	F	6/6	B	5.92446	8.3	66.2	-83.0	25.7	-74.7	285.9	353.0/18.0
37	F	3/6	C	2.99134	8.1	230.9	65.7	18.1	56.5	26.1	317.6/10.2
38	P	6/6	C	5.98283	3.9	291.1	46.4	24.9	46.2	36.8	29.8/11.3
39	F	6/6	C	5.91964	8.6	62.2	77.2	18.6	70.2	45.3	350.2/10.3
40	F	6/6	C	5.95907	6.1	122.2	77.4	13.9	70.8	42.7	350.2/10.3
41	F	6/6	C	5.98919	3.1	462.7	74.5	28.6	66.7	48.3	350.2/10.3
55	LS	5/6	C	4.95910	7.8	97.8	62.3	347.5	74.4	30.4	43.3/19.9
56	LS	6/6	C	5.97961	4.3	245.2	56.1	346.1	69.6	10.0	45.4/17.8
57	LS	6/6	C	5.96395	5.7	138.7	63.5	337.3	78.3	13.6	44.2/18.6
58	F	6/6	C	5.97481	4.8	198.5	42.9	279.7	63.6	293.1	350.8/22.8
59	F	5/6	C	4.99038	3.8	416.0	54.9	300.3	67.8	336.7	350.8/22.8
61	P	6/6	B	5.96711	5.5	152.0	-46.0	196.0	-33.6	203.0	323.7/14.9
62	PB	11/11	C	8.76777	24.3	4.5	61.3	296.2	38.1	316.2	252.2/28.1
63	F	6/6	B	5.98765	3.3	404.8	-21.4	196.2	-2.0	190.0	226.8/34.2
64	F	6/6	C	4.91248	11.4	45.7	83.1	281.7	77.8	15.6	314.2/14.5
65	F	5/6	B	4.98774	4.2	326.4	-47.5	197.7	-39.7	202.3	321.9/9.2
78	F	6/6	C	5.95350	6.5	107.5	51.6	41.6	61.7	199.4	122.0/65.7
79	F	5/6	C	4.99170	3.5	482.0	44.7	32.7	76.4	284.5	138.6/51.9
80	F	5/6	C	4.96849	6.8	126.9	47.7	42.8	79.5	250.5	138.6/51.9
81	F	6/6	C	5.99221	2.6	641.6	43.9	45.2	83.7	251.3	138.6/51.9
82	F	6/6	C	5.95416	6.4	109.1	61.1	35.5	65.6	222.7	128.8/53.2
83	F	6/6	C	5.97323	4.9	186.8	57.2	13.2	61.8	265.3	136.4/51.0
84	F	6/6	C	5.96178	5.9	130.8	58.7	28.7	67.3	250.5	136.4/51.0
85	F	6/6	C	5.99882	1.0	4254.6	64.6	22.1	64.5	59.2	40.3/17.2
86	F	6/6	C,F	5.98361	3.8	305.1	62.9	80.3	79.8	39.2	190.2/20.7
87	P	6/6	C,F	5.98460	3.7	324.6	69.2	75.1	61.0	5.1	232.5/30.5
88	F	5/6	C,F	4.99626	2.3	1068.3	68.4	86.2	64.1	55.5	268.3/13.9
89&12	F	10/12	C,F	3.99443	4.0	538.4	61.5	9.5	47.8	357.6	237.4/16.5
90	LS	6/6	C,F	5.98973	3.0	486.7	58.4	67.2	50.0	18.5	235.5/33.9
91	PC	5/6	C,F	4.95171	8.5	82.8	62.6	62.9	50.4	11.1	235.5/33.9
92	PC	6/6	C,F	5.95029	6.7	100.6	64.9	69.7	80.4	60.0	165.8/15.7
93	F	5/6	C,F	4.95408	8.2	87.1	71.0	54.6	71.0	331.3	191.3/25.1
94	LS	6/7	C,F	5.99716	1.6	1758.7	62.7	69.7	76.0	5.8	191.3/25.1
95	F	6/6	C,F	5.95006	6.7	100.1	71.4	51.5	82.2	322.8	164.9/20.1
96	F	6/6	C,F	5.97968	4.3	246.1	69.9	57.3	87.0	58.2	147.1/17.1
97	F	5/6	C,F	4.99409	2.9	676.3	57.1	64.1	84.9	116.7	145.9/30.2
98	F	5/6	C,F	4.99290	3.2	563.3	55.4	66.6	67.5	338.2	191.1/42.3
99	F	4/6	C,F	3.96196	10.4	78.9	51.3	104.7	73.2	245.0	179.1/26.8
100	F	5/6	C,F	4.97594	6.0	166.3	43.0	70.9	67.3	52.9	179.1/26.8
101	F	6/6	C,F	5.95256	6.6	105.4	71.1	68.9	66.1	351.1	218.0/27.9
102	F	6/6	C,F	5.96703	5.5	151.6	52.6	82.3	69.2	30.1	213.0/33.0

N is the number of stable samples in a locality out of total number collected, *I* is in situ inclination, *D* is in situ declination, *Ic* is tilt-corrected inclination, and *Dc* is tilt-corrected declination. Rock type code is Fs, lava flows; F, lava flow; P, pillow lavas; PB, pillow breccia; PC, pyroclastic deposit;

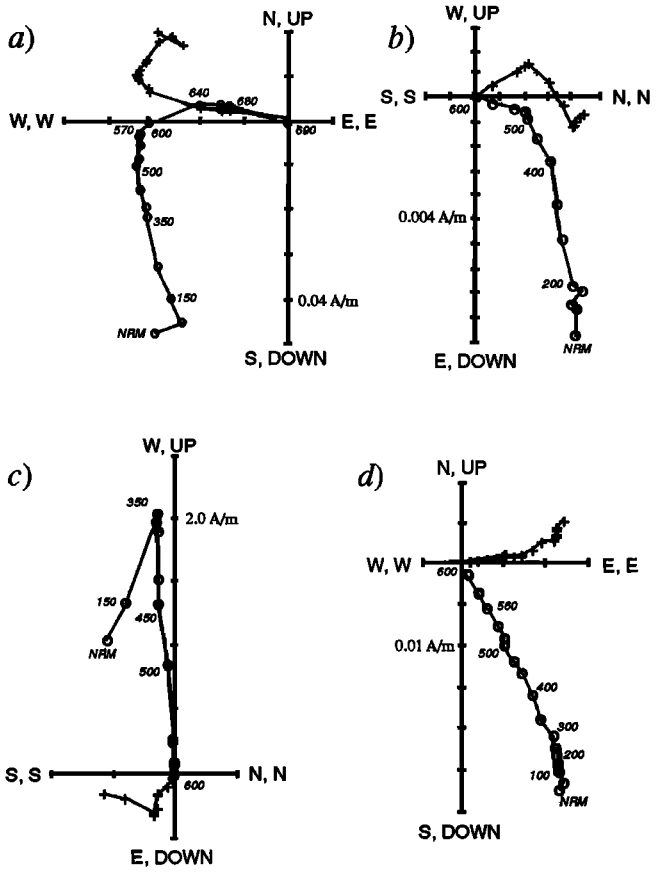


Fig. 3. Zijderveld diagrams showing in situ directions of typical samples with different components of magnetization of the Hound Island Volcanics. The plus and circle symbols denote the horizontal and vertical component of the remanence, respectively. The numbers next to the symbols give the temperature of the sample in degrees Celsius. (*a*) Component A (northwest and shallow) at temperatures between 300° and 500°C. Note component C (northeast and steeply down) is a secondary component, with the magnetization carried by magnetite. Sample is 86H026A from site 5. (*b*) Component A, with the magnetization carried by magnetite. Also note component C is a secondary component. Sample is 86H0002A from site 1. (*c*) Component B. Sample is 86H0031B from site 6. (*d*) Component C. Sample 87H0247A from site 34.

There is some evidence for both polarities of component A. Three sites have directions reversed to the mean tilt-corrected direction of component A and are unlike any other component of magnetization we find in the Keku Strait. One reversed direction is from a flow at site 3 at Cape Bendel (Figure 2). Only one sample was taken from this flow. The other two are site mean directions from one of the Keku Islets 14 km away (Figure 2). The Point Camden gabbro intrudes a portion of this island, but a positive baked-contact test indicates these two sites (28 and 29) were unaffected by the intrusion. The test was as follows. We drilled several sites from the gabbroic intrusion and away from its contact into the Hound Island Volcanics (sites 25 to 28; site 25 of the Point Camden gabbro is not shown on Figure 2 but is in the same location as site 26). The site in the Hound Island Volcanics closest to the pluton (site 26) does not have a significantly different direction from the site in the intrusion (site 25) at the 96% confidence level, suggesting that the gabbro overprinted this site (see *Haeussler et al.* [this issue], for directional data for site 25 in the Point Camden gabbro). However, the site furthest away from the contact (site 28) has a very different high-temperature direction, which is reversed relative to component A. This direction is not similar to any other component we observed in the Keku Strait, and thus they constitute a positive baked-contact test. Site 27 has a direction that lies on a great circle between sites 26 and 28, indicating it was partially remagnetized. Site 29, which lies a comparable distance from the contact of the Point Camden gabbro as site 28, has a reversed direction similar to that of site 28.

A number of samples with the component B direction (discussed in the next section) did not decay univectorally toward the origin of Zijderveld diagrams, which suggests an unresolved component of magnetization. These samples were consistently heading toward a northwest and shallow direction, that is, toward component A. This unresolved component was crudely defined using the principal-component (great-circle) method of *Kirschvink* [1980]. This method finds the great circle that the demagnetization points for one sample lie upon. The poles to the great circles from all samples should define a plane. One of the poles to this plane gives the direction of the unresolved component, the pole toward which the demagnetization path follows on its great circle. The great circles defined by our samples did not converge enough to justify using them in the absence of any known directions. For sites where at least one sample reached a stable component A endpoint, we employed the method of *McFadden and McElhinny* [1988] in order to combine the known direction(s) with the great circles and calculate the best mean direction and confidence limit. We used this procedure with data from sites 4, 6, 95, and 96, as denoted in Table 2.

Rock magnetism. The intensity of the remanence after component A is isolated is often less than 10% of the NRM. The magnetization of component A rocks resides in both hematite and magnetite. The presence of titanium-poor magnetite is indicated by unblocking temperatures between 540° and 580°C (Figure 3*b*) and by plateaus between 150 and 200 mT on IRM acquisition curves [*Haeussler*, 1991]. Hematite is most likely the carrier of the remanence in about a third of these sites, as indicated by unblocking temperatures from 600° to 680°C (Figure 3*a*). IRM acquisition curves for some samples also indicate that hematite is present.

Reflected light microscopy. We examined 31 thin sections of the Hound Island Volcanics with a reflected light microscope at 1000x in oil. We commonly observed, in rocks with component A, exsolution textures within titanomagnetites typical of high-temperature oxidation. These rocks are highly oxidized, characteristic of *Haggerty*'s [1976] C3 and C4 stages. In rocks with the C3 stage, we observed ilmenite lamellae exsolved along three lattice planes forming a trellis pattern within the magnetite host. This exsolution tends to produce nearly pure magnetite, consistent with the rock magnetic evidence for titanium-poor titanomagnetite. In rocks with the higher C4 stage of oxidation, we also observed ilmenite altering to rutile and hematite. The hematite is also present as a mass of very fine grained crystals rimming exsolved titanomagnetite or solitary ilmenite crystals. Because the hematite and rutile are not associated with fractures, but rather are usually found in association with ilmenite or titanomagnetite, we infer that the hematite was produced during, or soon after, the time of eruption.

None of the rocks that were oxidized to Haggerty's lower C1 or C2 stage of oxidation showed component A. Some of these rocks were ex-

LS, limestone; and S, clastic sediments. Comment code is B, component B; C, component C; F, directions from Cape Bendel and Turnabout Island used for final calculation of mean; U, unknown component. 6GC's means six great circles were used in calculating the mean direction. The R, α_{95}, and k listed are for the in situ directions. The mean of the component C in situ virtual geomagnetic poles (VGPs) for the Hound Island Volcanics, with sites 29 and 62 ommitted (see text for explanaition), is Lat = 66.3°, Long = 313.0°, α_{95} = 7.3°, K = 7.7, R = 48.90205, N = 56. The mean of the in situ VGPs for both the Hound Island Volcanics component C (with sites 29 and 62 ommitted) and the Permian Pybus Formation (Table 4) is Lat = 67.1°, Long = 314.4°, A_{95} = 6.4°, K = 8.7, R = 56.73829, N = 64. The average site latitude and longitude is 57.1°N, 226.4°E.

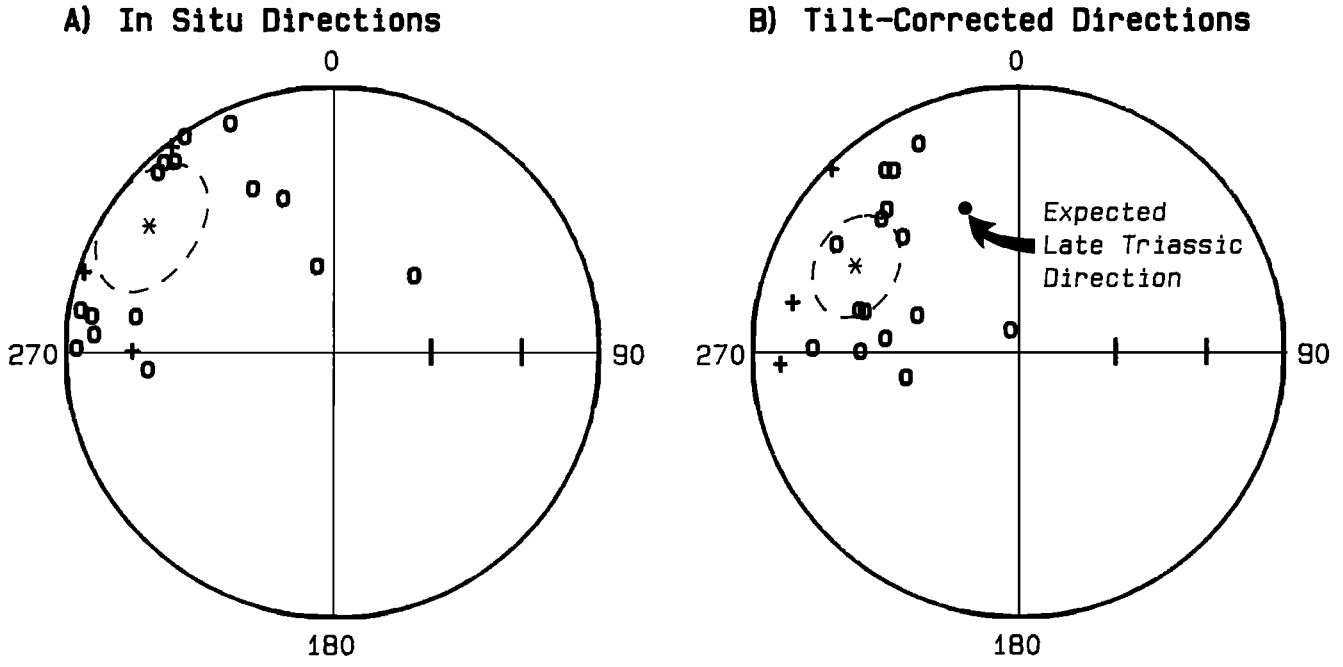


Fig. 4. Fold test for component A in the Hound Island Volcanics. Lower hemisphere directions are pluses and upper hemisphere directions are circles. These are the site or grouped mean directions as given in Table 2. We interpret three directions as reversed and they are inverted through the origin. (*a*) In situ directions. (*b*) tilt-corrected directions. These data pass a tilt test [*McElhinny*, 1964] at 82% confidence and a more rigorous tilt test [*McFadden*, in press] at 99% confidence. The large dot shows the expected Late Triassic direction for North America (Dec = 338.6°, Inc = 48.3°) using the combined Manicouagan-Chinle pole favored by *May and Butler* [1986].

solved on only one lattice plane, whereas others were not exsolved but showed granulation texture. Often, regardless of what component of magnetization the rocks record, sphene, identified optically and by electron microprobe, has replaced the ilmenite lamellae, the titanium-poor titanomagnetite, or both, in the exsolved titanomagnetite crystals. Sometimes the sphene so beautifully pseudomorphed the exsolved ilmenite and magnetite that the degree of oxidation could still be discerned, but not always. We believe that this "sphenetization" of the titanomagnetite and ilmenite crystals occurred soon after deposition of the rocks, during the sub-greenschist-facies metamorphism of the Hound Island Volcanics that occurred soon after eruption of the lavas onto the seafloor. *Haggerty* [1976, pp. Hg-119-122 and Hg-129-134] reports that "sphenetization is relatively common in regional low-grade metamorphic terrains...".

Discussion. Several lines of evidence point to a primary origin of component A magnetization. First, and most important, is that the directions pass a fold test, demonstrating that the remanence predates Cretaceous folding and metamorphism. Second, this component resides in both hematite and magnetite, which attests to its stability and indicates that both minerals were magnetized at approximately the same time. Third, this direction of remanence, when present, was always the highest-temperature component of magnetization, indicating it is the most stable. Component B (observed in the majority of *Hillhouse and Grommé*'s [1980] sites) is found only at temperatures lower than those of component A. Finally, we may observe both polarities of component A.

The presence of the primary component (A) in lava flows and pyroclastic deposits, never in pillow lavas or pillow breccias, is consistent with observations from submarine seafloor basalts. These studies show that pillow lavas cool too rapidly to allow exsolution of titanomagnetite grains. Thus the magnetite grains are titanium-rich, with low Curie temperatures, which permits the rocks to be easily overprinted at a later time. In contrast, sheeted dikes, like the lava flows in the Hound Island Volcanics, cool slowly, allowing high-temperature oxidation-exsolution to occur, which produces a stable magnetization in titanium-poor titanomagnetite with high Curie temperatures.

The intensity of NRM of rocks with component A (between 10 and 10^{-3} A/m)are considerably less than those of pristine seafloor basalts, and component A is often isolated at ≤ 10% of the NRM. We believe this effect is caused by the sphenetization of the titanomagnetite grains, which appears to decrease their number and thereby reduce the NRM.

We eliminated site 99 (Figure 2 and Table 2) from the final calculations. This site has a direction unlike any other in the Keku Strait (southwest and shallow downward), yet the high-temperature component resides in hematite and behaves most like those samples with a component A direction. Perhaps this site records a transitional direction.

The directions in the Hound Island Volcanics are more scattered than would be expected if secular variation during Triassic time were similar to that during the last 5 m.y. Angular standard deviation S for our sites is 29°, whereas the expected value at the inferred paleolatitude is 13° [*McFadden et al.*, 1991]. Larger dispersion is not unexpected, given the uncertainties inherent in the structural corrections and the probability of incomplete removal of secondary components in some samples.

We conclude that component A is the primary component of magnetization in the Hound Island Volcanics. The presence of this component only in highly oxidized lava flows and pyroclastic deposits, and not in less oxidized flows or in pillow lavas, which are easy to paleomagnetically reset, argues for its acquisition upon cooling of the rocks. Another possibility is that the remanence was acquired within a geologically short time after eruption of the lavas by hydrothermal alteration on the seafloor. We consider this option somewhat less likely because, if it were true, we might expect to find component A in all lithologies, in contradiction to our finding component A in rocks with highly oxidized titanomagnetite grains.

Component B: Southerly and Steeply Upward Direction

Nine of the 70 sites in the Hound Island Volcanics show a distinctive southerly and steeply upward to vertical component (component B). We found this component only in samples obtained from portions of Hound Island and Kuiu Island (Figure 2). We did not observe this direction as a

secondary component on component A or C, or vice versa. Several lines of evidence indicate that magnetite is the carrier of the remanence. Blocking temperatures of ≤ 580°C (Figure 3*d*) and IRM acquisition curves with plateaus around 150-200 mT indicate the remanence resides in magnetite [*Haeussler*, 1991]. Also, the thermomagnetic curves of *Hillhouse and Grommé* [1980] suggest that titanium-poor titanomagnetite is the dominant magnetic mineral in flows with this direction on Hound Island.

In contrast, reflected light microscopy of rocks with component B indicates that they are not highly oxidized (*Haggerty*'s [1976] C1 and C2 stages). We observed titanomagnetite grains that were unexsolved with granulation texture or were exsolved on only one lattice plane. Some of the titanomagnetite grains were partially to extensively replaced, but not pseudomorphed, by sphene.

A fold test with the component B directions is inconclusive; the precision parameter k is essentially unchanged after unfolding (it goes from 6.8 to 7.1). However, only nine sites exhibit this component and there is only minor variation in the structural correction between sites, so that a diagnostic result from the fold test would not be expected.

We conclude that this component is an overprint caused by the intrusion of the Point Camden gabbro, because its orientation is similar to reversed polarity directions in nearby sites of the Point Camden gabbro. *Haeussler et al.* [this issue] have studied the paleomagnetism of these 23-Ma intrusions, which intrude the Hound Island Volcanics in the Keku Strait. The Point Camden gabbro has both normal and reversed polarity directions that average to the expected direction for North America for early Miocene time. Reversed directions from the Hound Island Volcanics on Hound Island (sites 32-36) are not distinct at 95% confidence from those directions at the closest outcrops of the Point Camden gabbro (sites 50-53; see *Haeussler et al.* [this issue], for directional data from the Point Camden gabbro) (see Figure 5). Also, the south-southwest and upward directions in the Hound Island Volcanics on Kuiu Island (sites 16, 61, and 63) are very similar to the mean direction of site 54 in the Point Camden gabbro. Just across Kadak Bay, site 66 is in a thick (≈ 100 m) dike of Point Camden gabbro, which also has a southerly and upward direction not very different from those in the Hound Island Volcanics. Because the directions from both units are similar, we believe the Point Camden gabbro overprinted the Hound Island Volcanics on Hound Island and portions of Kuiu Island 23 m.y. ago. These data suggest the Point Camden gabbro may extend much closer to Hound Island than surficial mapping indicates.

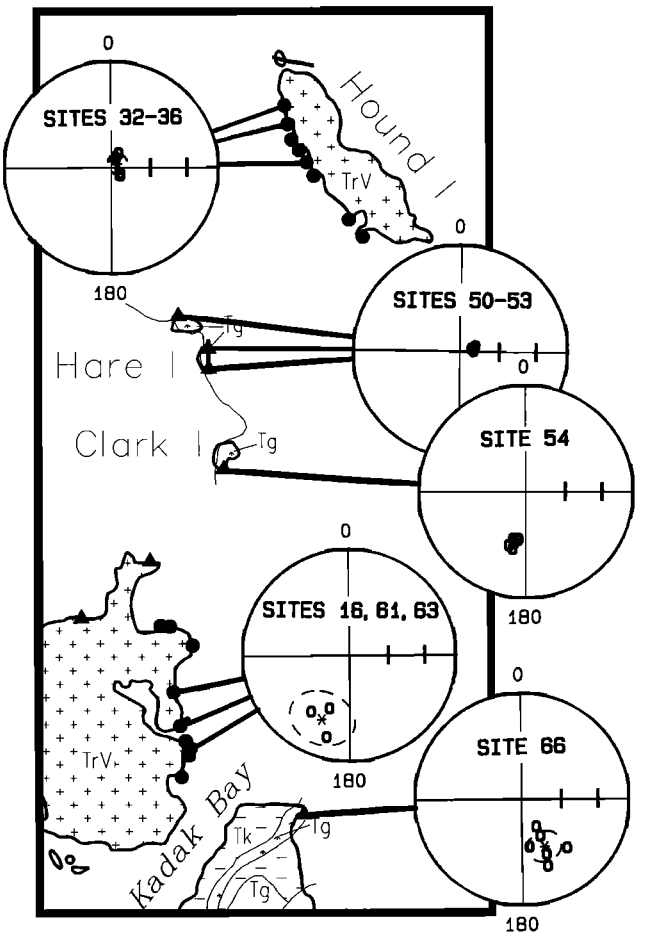


Fig. 5. Geologic map of Hound Island and adjacent islands in the Keku Strait and a portion of Kuiu Island. Map symbols used are the same as in Figure 2. Also shown are equal-area stereonets of upper hemisphere (component B) site mean directions from the Hound Island Volcanics (sites shown with dots) and site mean directions from the Point Camden gabbro (sites shown with triangles), which have reversed directions. The similarity of the directions in the Hound Island Volcanics to those in the Point Camden gabbro leads us to conclude that the Point Camden gabbro overprinted a portion of the Hound Island Volcanics. Dimensions of the map are 3.3 x 6.6 miles.

Component C: Northeast and Steeply Downward Direction

We observed north-to-northeasterly and steeply down components of magnetization in 56 out of 70 sites in the Hound Island Volcanics (component C) and in all eight sites of the Permian Pybus Formation. The great number of similar normal polarity directions in units with two different ages suggests it is an overprint. In addition, a regional fold test with all sites that show a north-to-northeasterly and down direction demonstrates this magnetization was acquired after folding (Figure 6). Moreover, we found this component of magnetization in all Late Triassic lithologies: lava flows, pillow lavas, broken pillow breccias, volcaniclastic rocks, and limestones.

Additional tests support the conclusion that the north-to-northeast and down direction is an overprint. We performed a conglomerate test on separate clasts within a broken-pillow breccia on Hound Island (site 31) and on Kuiu Island (site 62; see Figure 2). All sample directions are similar within site 31: northeast and down (Figure 7*a*).This demonstrates that the magnetization was acquired after the clasts reached their present orientation. Directions in clasts at site 62 are more scattered (Figure 7*b*) but, nevertheless, are broadly consistent with the interpretation that these rocks are overprinted.

At four localities we drilled successive lava flows, which had siltstone sediment horizons between them, for what we will refer to as a "secular variation test" (Figure 8). These siltstone beds are homogeneous and lack the graded bedding of turbidites, implying slow deposition over a long enough period that the interbedded flows could have recorded significant secular variation if their magnetization were primary. However, the directions between these flows are virtually identical. Three of these localities had north-to-northeast and down directions (sites 39-41 on Hound Island and sites 79-81 and 83 and 84 on Hamilton Island; see Figure 2) and the fourth had a south and upward direction (component B) (sites 32-33 on Hound Island; see Figure 2). The site mean directions from sites 39, 40, and 41 have a combined angular standard deviation (S) of 2.4°. Also, sites 79, 80, and 81 have an S of 5.1°. Moreover, sites 83 and 84 have a low S of 5.9°. Finally, the site mean directions from sites 32 and 33 on Hound Island have an S of 2.8°. In contrast, the expected value of the angular standard deviation, based on models of secular variation for the last 5 m.y. is 18. We contend it is more likely that the small amount of scatter in successive flows in these four cases is due to overprinting rather than the chance that overlying flows recorded exactly the same direction in the secular variation of the Earth's magnetic field.

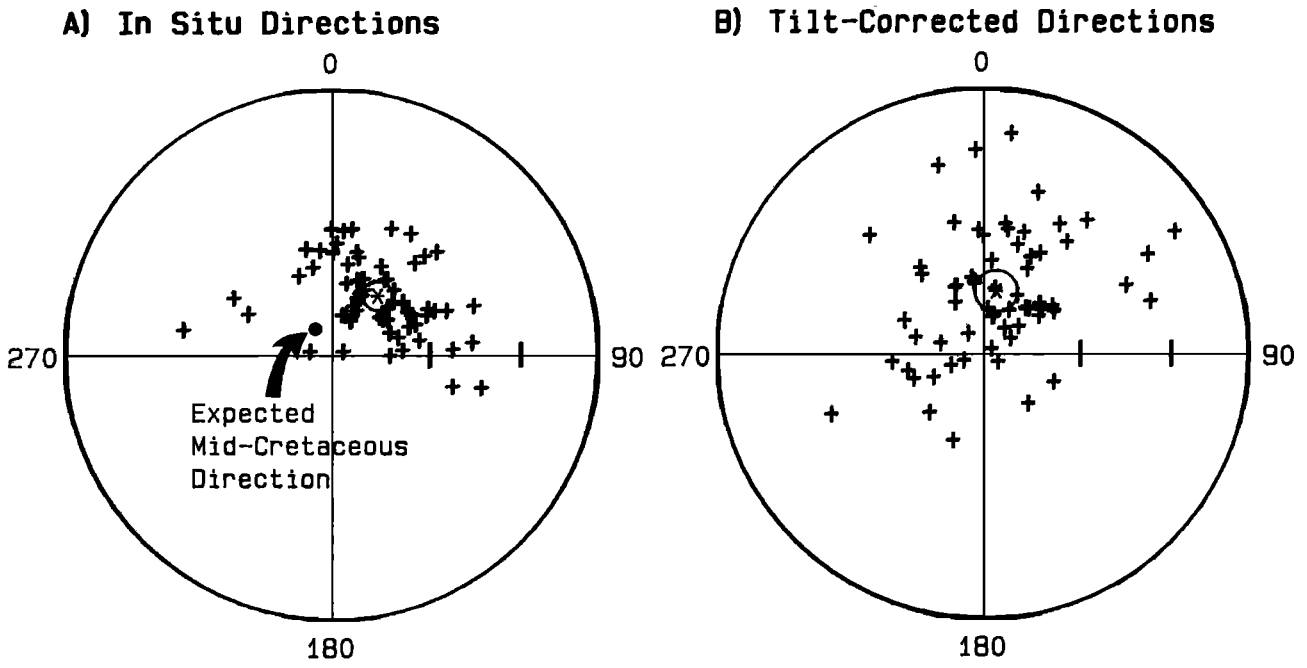


Fig. 6. Fold test of all northerly and down directions (component C) in the Hound Island Volcanics and the Permian Pybus Formation. Fisher's k decreases from 15.5 to 7.7 after unfolding with $N = 66$, demonstrating these fail a fold test. The mean direction before tilt correction is Dec = 35.1°, Inc = 67.3°, $\alpha_{95} = 4.6°$, $k = 15.5$, $N = 66$. The expected early Late Cretaceous direction is Dec = 329.4°, Inc = 80.2° [*Globerman and Irving*, 1988].

Blocking temperatures for rocks with component C are generally ≤ 600°C and suggest the magnetic carrier is magnetite (Figure 3*c*). We have measured lateral temperature variations of up to 30°C in our oven, and thus we attribute blocking temperatures in the range from 580° to 610°C to these variations. IRM acquisition curves only indicate the presence of magnetite within the samples. In addition, the thermomagnetic curves of *Hillhouse and Grommé* [1980] on samples from Hamilton Island (Figure 2) indicate that this component resides in titanium-poor titanomagnetite (T_c = 540°C). However, decay of remanence curves during thermal demagnetization generally show a wide range in unblocking temperatures, suggesting a higher titanium content. These samples are in contrast to those which carry a primary magnetization (component

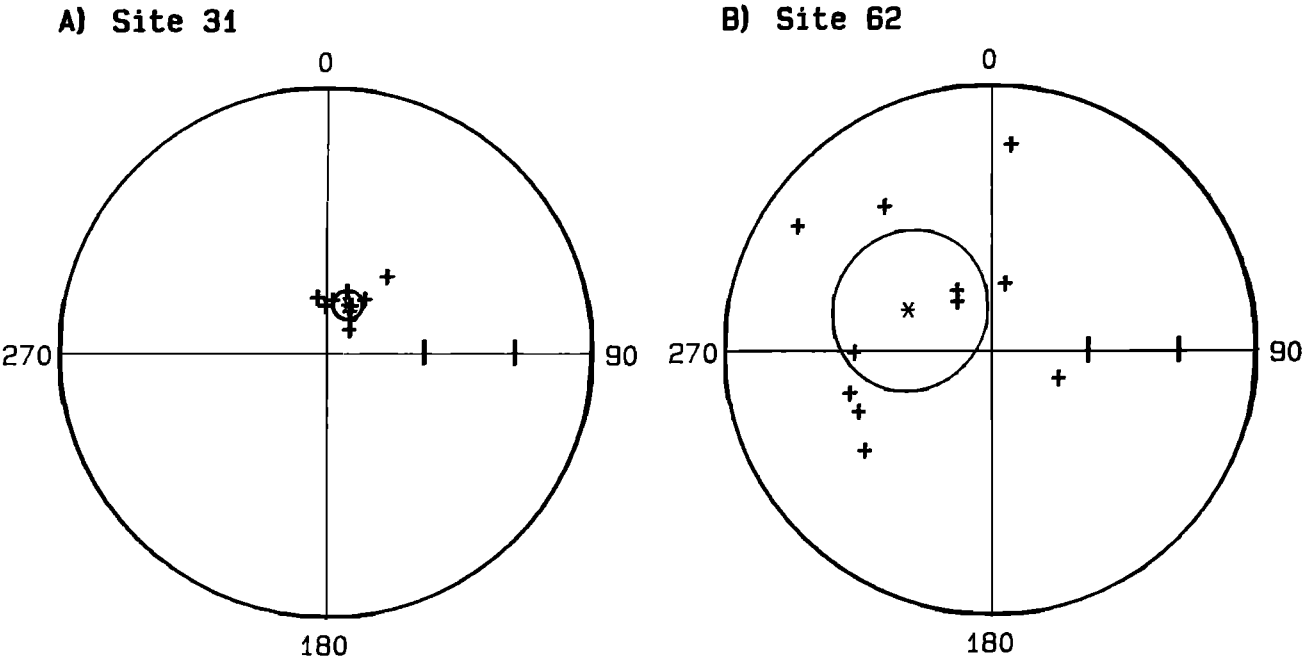


Fig. 7. Conglomerate test on a broken pillow breccia at (*a*) site 31 and (*b*) site 62 of the Hound Island Volcanics (site locations shown in Figure 2). The stereonets show the in situ best fit sample directions, with the mean direction and α_{95} circle of confidence.

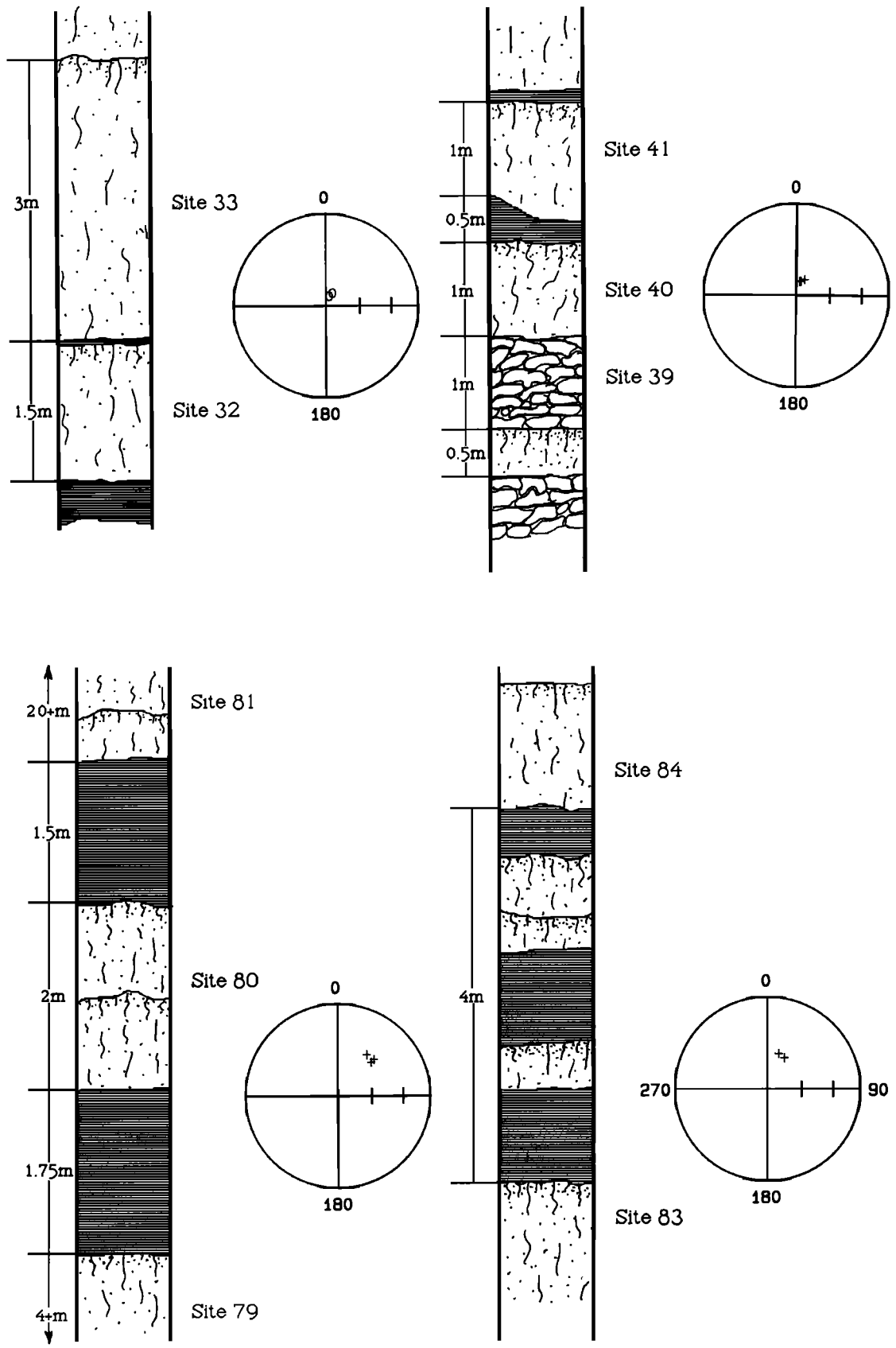


Fig. 8. "Secular variation" test in the Hound Island Volcanics, showing stratigraphic columns and paleomagnetic directions from four localities (sites 32-33, 39-41, 79-81, 83-84; see Figure 2 for site locations). At these four localities, there is almost no variation in paleomagnetic directions even though there are sedimentary interbeds between lava flows, suggesting that these rocks are overprinted. The plus and circle symbols indicate lower and upper hemisphere directions, respectively.

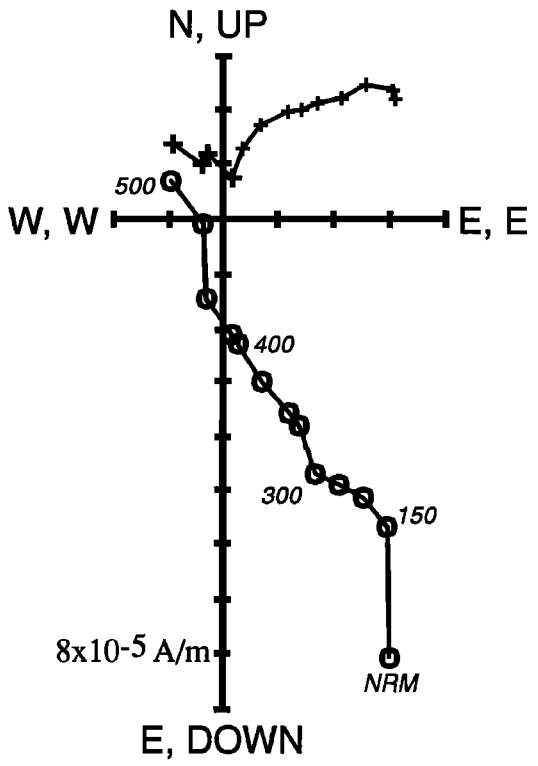


Fig. 9. Vector diagram from the Permian Pybus Formation. The plus and circle symbols denote the horizontal and vertical component of the remanence, respectively. The numbers next to the symbols give the temperature of the sample in degrees Celsius. Note that the remanence does not decay directly to the origin, suggesting an unremoved component. Also, at the last step in the demagnetization sequence, the intensity increases, indicating alteration of the magnetic mineralogy, typical of these samples. Most samples have a greater increase in intensity than this one. Sample 87H0313A is from site 45. In situ directions.

A), which resides in magnetite with high unblocking temperatures. These magnetite grains often unblock over a narrow temperature range of less than 20°C.

As mentioned previously, reflected-light microscopy of the rocks that have component C showed only lower stages of oxidation-exsolution (*Haggerty's* [1976] C1 and C2 stages). We observed within titanomagnetite grains either no exsolution lamellae or exsolution on only one plane. Sphenetization is generally more extensive in these rocks than in those with component A. This relationship between overprinting and lower oxidation state is easily understood. Without the high degree of oxidation, the higher titanium content of the titanomagnetite lowers the Curie temperature, which causes the rocks to be more susceptible to thermal remagnetization. Conversely, the highly oxidized rocks have a low titanium content, thus they have high Curie temperatures, and therefore they are less prone to thermal remagnetization.

To summarize, we interpret component C as a regional overprint for several reasons: (1) it is often found as a secondary component on the primary component (A); (2) it fails a fold test; (3) it fails a conglomerate test; (4) it fails three "secular variation" tests, and (5) the magnetic mineralogy of samples containing component C are less oxidized than those with component A. We believe the lower oxidation state, and thus the lower Curie temperature, of titanomagnetite grains in rocks with component C led to thermal overprinting. We do not believe component C represents a recent-field overprint because the direction is shallower and more easterly.

For our final calculations involving the component C direction we excluded sites 29 and 62. In site 29 the secondary components of magnetization (component C?) are poorly constrained and do not show consistent directions between most samples. The best fit directions for the samples in site 62 are very scattered as shown on Figure 7*b*, and thus we do not consider the mean direction to be very reliable. Sites 58 and 59 have unusual west-northwesterly declinations, yet the characteristic component in all samples from these two sites is very well defined. Therefore we find no convincing reason to exclude these directions. In addition, because statistical distributions are expected to have "outliers," directions such as these are expected. Finally, because the population is large, the inclusion or exclusion of these two sites has little effect on the final result.

Permian Pybus Formation Dolomite

The Permian Pybus Formation crops out in the northern Keku Strait on the Keku Islets and on northern Kuiu Island (Figure 2). The Pybus formation consists of limestone, dolomite, and chert. We gathered samples at eight sites on the Keku Islets: two groups of four sites, each group with opposite dips. Thermal demagnetization of the samples was better at isolating the remanence directions than AF demagnetization. However, between 480° and 520°C, near the end of the demagnetization sequence, there was usually a profound jump in the measured intensity of remanence, indicating a change in magnetic mineralogy. Figure 9 shows a Zijderveld diagram from a sample in which there was only a modest increase in the intensity of the remanence at the higher-temperature demagnetization steps. IRM acquisition curves show much higher saturation remanences in heated samples than in unheated ones, also indicating a change in magnetic mineralogy [*Haeussler*, 1991]. This jump in intensity may be related to formation of magnetite from an iron sulfide mineral.

After correcting for the dip of the bedding, it is clear that the characteristic directions fail a fold test (Figure 10 and Table 4): k decreases from 135 to 6, demonstrating a postfolding overprint. The similarity of this overprint direction to component C in the Hound Island Volcanics leads us to believe the cause of the overprinting is the same. We consider it less likely that the Point Camden gabbro overprinted these rocks, because there are no large intrusions nearby.

Age of Overprinting

We believe that the reversed directions in the Hound Island Volcanics on Hound Island and Kuiu Island are the result of a thermal overprint caused by intrusion of the Tertiary gabbro at 23 Ma. However, the age of the normal-polarity overprint (component C) is more difficult to determine. It seems likely the Late Cretaceous and early Tertiary orogenic event in southeastern Alaska caused the overprinting of the Hound Island Volcanics. We attempted to provide better constraints by using $^{40}Ar/^{39}Ar$ geochronology.

$^{40}Ar/^{39}Ar$ Geochronology

A more satisfying method of constraining the age of the paleomagnetic overprint would be to date the potassium-rich metamorphic minerals identified by electron probe analyses of the Hound Island Volcanics. In principal the $^{40}Ar/^{39}Ar$ laser probe is ideal for dating the fine-grained ferric biotite, which occurs around the margins of pyroxenes, and also abundant sericite, which is within relict plagioclase laths of some lava flows. Because the closure temperatures for argon diffusion in micas are very low, the $^{40}Ar/^{39}Ar$ isotopic ages should reflect the age of the latest metamorphic event. We analyzed those samples with the largest sericite crystals (1-20 μm), which were from Turnabout Island and on Kupreanof Island at Cape Bendel (Figure 2).

Sample preparation for analysis with the Princeton University $^{40}Ar/^{39}Ar$ laser microprobe involved making circular polished thick

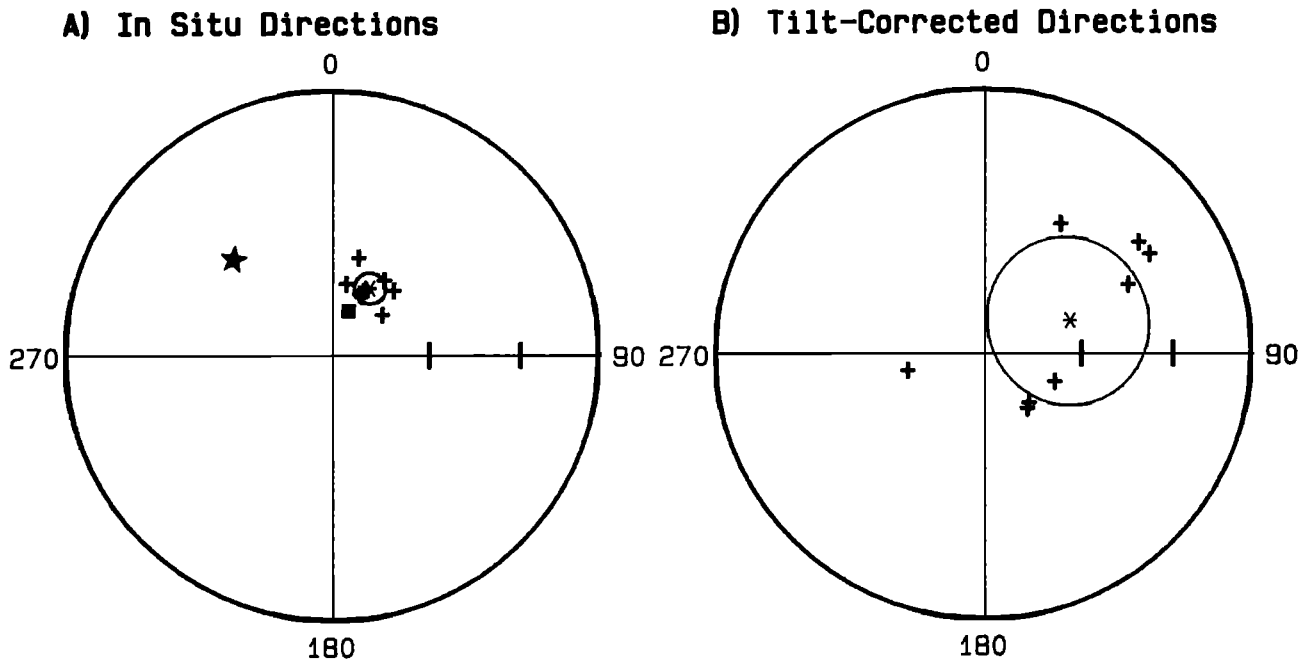


Fig. 10. Fold test of site mean directions from the Pybus Formation: (*a*) in situ directions and (*b*) tilt-corrected directions. Fisher's *k* decreases from 135.4 to 5.7 after unfolding with $N = 8$, which demonstrates the data fail a fold test. Also shown are the present-day field direction with a square, the expected early Late Cretaceous direction with a dot [*Globerman and Irving*, 1988], and the expected Lower Permian direction for North America with a star. The Permian direction is a mean of those favored by *Van der Voo* [1989].

sections, 200 μm thick by 3 mm in diameter. The samples were cut from unheated paleomagnetic cores, one section from sample 21 at site 4, three from sample 27 at site 5, and two from sample 29 at site 5. The samples were individually wrapped in aluminum foil packets and stacked into an aluminum irradiation can with flux monitors (standards) of Mmhb-1 homblende (520 ± 1.7 Ma, [*Samson and Alexander*, 1987]) evenly dispersed to measure any flux gradients. Laser "spot" dates were obtained by focusing the laser beam (12 μm diameter) and instantaneously volatilizing or melting tiny volumes of the thick section. Detailed analytical procedures are given by *Lee et al.* [1990]. The $^{37}Ar/^{39}Ar$ values are related to Ca/K by the expression $Ca/K = 1.82(^{37}Ar/^{39}Ar)$ [*Onstott and Peacock*, 1987].

In practice, it was not possible to analyze individual sericite and biotite crystals with the laser probe because the amount of ^{36}Ar released from a single grain was too small for accurate atmospheric correction. We therefore fused small regions of the sample (0.1-0.5 mm in diameter). Each analysis was dominated by the most abundant mineral species with the most argon; however, inspection of the samples after the analyses were completed indicated the ferric biotite was preferentially fused compared to the sericite within the plagioclase. This is because the absorption coefficient for 1.04 μm radiation of the sericite and enclosing plagioclase was too low, and complete fusion could only be obtained by laser heating of mafic inclusions or adjacent mafic grains.

The 37 analyses we obtained yield apparent ages ranging from 55.7 to 249.9 Ma (Table 5). On an isotope correlation diagram [*Turner et al.*, 1971], 32 of these analyses fall within a triangular field (see Figure 11). This field is demarcated by the $^{39}Ar/^{40}Ar$ axis and two chords. One chord, defined by four points, connects an $^{39}Ar/^{40}Ar$ ratio corresponding to an age of 197 ± 7 Ma with an $^{36}Ar/^{40}Ar$ corresponding to an $^{40}Ar/^{36}Ar$ ratio of 363.7 ± 6.0. The second chord, defined by 13 points, connects an $^{39}Ar/^{40}Ar$ ratio corresponding to an age of 108 ± 11 Ma, with an $^{36}Ar/^{40}Ar$ corresponding to an $^{40}Ar/^{36}Ar$ ratio of 381.5 ± 6.6. The $^{40}Ar/^{36}Ar$ ratio at the apex of the triangular field is greater than the accepted ratio for atmospheric argon of 295.5 but far less than $^{40}Ar/^{36}Ar$ ratios determined from young MORB glass, which are greater than 10,000 [*Fischer*, 1986].

TABLE 4. Paleomagnetic Data for the Permian Pybus Formation

Site	*N*	*R*	α_{95}	*k*	*I*	*D*	*Ic*	*Dc*	Strike/Dip
42	3/6	2.92756	23.9	27.6	58.7	15.1	42.7	30.5	331.0/20.1
43	6/6	5.95027	6.7	100.5	70.1	51.1	39.9	64.3	344.4/31.3
44	5/6	4.92424	10.6	52.8	61.7	34.8	29.4	54.1	344.4/36.5
45	6/6	5.85084	11.7	33.5	62.4	43.6	28.4	58.8	344.4/36.5
46	5/6	4.97570	6.0	164.6	69.7	26.7	68.5	141.4	83.1/35.7
47	5/6	4.95020	8.6	80.3	67.9	25.9	69.5	137.5	83.1/35.7
48	6/6	5.89967	9.6	49.8	69.2	24.0	66.6	111.4	64.3/31.4
49	5/6	4.94768	8.8	76.5	67.5	11.0	66.2	257.5	135.3/39.4
Mean	8	7.9483	4.8	135.4	66.4	28.7	62.0	68.7	

N is the number of stable samples in a locality out of total number collected, *I* is in situ inclination, *D* is in situ declination, *Ic* is tilt-corrected inclination, and *Dc* is tilt-corrected declination. The *R*, α_{95}, and *k* listed are for the in situ directions. The Fisher mean of the in situ virtual geomagnetic poles gives a paleomagnetic pole of Lat = 71.6°N, Long = 326.1°E, $A_{95} = 6.8°$, $K = 67.1$, $R = 7.89561$, $N = 8$. The average site latitude and longitude is 56.9°N, 226.0°E.

TABLE 5. $^{40}Ar/^{39}Ar$ Laser Probe Data From the Hound Island Volcanics

	$^{36}Ar/^{39}Ar$		$^{37}Ar/^{39}Ar$		$^{38}Ar/^{39}Ar$		$^{40*}Ar/^{39}Ar$			
Sample	$^{36}Ar/^{39}Ar$	absolute error	$^{37}Ar/^{39}Ar$	absolute error	$^{38}Ar/^{39}Ar$	absolute error	$^{40*}Ar/^{39}Ar$	absolute error	Moles ^{39}Ar	Apparent Age, Ma
21	7.35626170E-02	1.78066470E-03	2.58148690	3.67184036E-02	7.75720008E-02	2.77211950E-03	18.21383760	0.829066205	5.69140325E-16	160.38 ± 8.9
21	9.68962312E-02	3.73894737E-03	2.83428695	4.03005522E-02	6.13841597E-02	1.86764899E-03	20.78237830	1.438034030	5.67034165E-16	181.89 ± 13.49
21	6.87343618E-02	8.98007531E-04	3.21942187	4.28110918E-02	9.43974551E-02	1.65822262E-03	16.81593130	0.536777045	3.33663232E-16	148.56 ± 6.85
21	4.64941621E-02	5.18027361E-04	4.58967622	4.90015308E-02	6.73657462E-02	6.87041974E-04	16.07892220	0.506050284	7.48506279E-16	142.3 ± 6.53
21	9.50938089E-02	1.33242003E-03	2.05318603	3.01613930E-02	7.01521940E-02	1.04394869E-03	17.63659530	0.793422141	3.88315581E-16	155.51 ± 8.58
21	4.90019234E-02	5.25573454E-04	3.37581575	3.31993974E-02	7.10743915E-02	1.15347199E-03	16.21585240	0.508035155	7.79461346E-16	143.47 ± 6.57
21	5.30877316E-02	6.62506197E-04	1.04378285	1.24257597E-02	3.97347420E-02	8.88810796E-04	14.93643670	0.507890224	5.41238777E-16	132.55 ± 6.32
21	6.95105991E-02	8.70629231E-04	9.09249933	1.00686856E-01	8.23363244E-02	2.19482565E-03	16.96708940	0.404604100	3.02914542E-16	149.84 ± 6.20
27	8.76936768E-02	1.80845167E-03	2.42134708	3.19776144E-02	7.70208456E-02	1.46057170E-03	25.32510770	1.066685380	7.02872012E-16	219.32 ± 11.43
27	8.70186905E-02	1.79502949E-03	1.39233765	1.81054719E-02	8.46189950E-02	2.20606225E-03	23.91798990	1.108513820	7.16215131E-16	207.81 ± 11.51
27	7.24603279E-02	1.62746352E-03	1.45072794	2.20298457E-02	7.69510877E-02	1.54667687E-03	23.60155610	1.007964550	8.58251008E-16	205.21 ± 10.82
27	4.22171659E-02	5.55777274E-04	4.21804379	5.48250499E-02	4.99302986E-02	9.45236433E-04	25.09783980	0.715815509	7.77081642E-16	217.47 ± 9.40
27	4.49609878E-02	8.89277458E-04	1.89743189	2.91684785E-02	5.97165792E-02	1.44313417E-03	20.37677860	0.643463299	6.06185824E-16	178.51 ± 8.13
27	5.33698040E-02	1.17044103E-03	1.74242403	2.21474705E-02	7.60185225E-02	2.64682542E-03	18.22484650	0.426231231	1.70547609E-16	160.47 ± 6.58
27	7.20811848E-02	9.16495593E-04	2.05681500	2.65280452E-02	8.08540330E-02	1.48573461E-03	21.20356360	0.342924831	6.38445662E-16	185.4 ± 6.94
27	4.09638184E-02	2.99736411E-04	3.44557264	1.99710475E-02	5.53385031E-02	4.70342950E-04	6.14697463	0.357110461	4.42920980E-15	55.74 ± 3.75
27	6.56535950E-02	9.75147045E-04	1.59952864	1.99485996E-02	6.77570213E-02	1.28997044E-03	20.81131350	0.582892863	2.51023659E-16	182.13 ± 7.89
27	9.26575340E-02	1.32253964E-03	2.18568854	2.85557488E-02	7.65685032E-02	2.55099670E-03	21.07712400	0.642098067	1.66304923E-16	184.34 ± 8.25
27	9.59026169E-02	1.38290205E-03	2.47085815	3.40536113E-02	1.04160373E-01	2.40681245E-03	20.91965180	0.433718980	3.57770305E-16	183.04 ± 7.22
27	9.41876901E-02	4.20999656E-03	1.66471732	4.84252477E-02	8.86721261E-02	7.22322414E-03	29.11237130	1.437923300	6.05583010E-17	249.94 ± 14.25
27	8.28799887E-02	1.49058457E-03	4.52043277	7.61792360E-02	6.73037908E-02	9.03920701E-04	24.69221320	0.905135071	5.28565352E-16	214.15 ± 10.36
27	3.24506468E-02	4.05638023E-04	4.31470376	4.78415446E-02	4.73171963E-02	7.43251042E-04	24.16236570	0.628478127	8.07718780E-16	209.82 ± 8.78
27	6.41041470E-02	9.94863235E-04	2.98167532	4.17037748E-02	6.91843123E-02	1.34140533E-03	23.65386940	0.817800974	5.64562091E-16	205.64 ± 9.69
27	7.96971450E-02	1.28858193E-03	3.61623928	5.48266094E-02	7.93894789E-02	3.08810977E-03	21.86185300	0.964724035	4.16836778E-16	190.86 ± 10.30
27	5.15646339E-02	8.73713711E-04	4.18170819	6.68205590E-02	5.96772888E-02	1.80298501E-03	22.22049420	0.619404910	5.86849189E-16	193.83 ± 8.36
29	1.87508491E-02	1.45681757E-03	7.34703782	7.72269417E-02	3.58161388E-02	6.95330355E-04	19.84442670	0.652055467	2.40227549E-15	174.07 ± 8.08
29	1.60711157E-02	3.58282563E-04	4.95255407	4.38874210E-02	3.41762530E-02	5.72911993E-04	23.32510050	0.545233384	1.95160137E-15	202.94 ± 8.23
29	5.31696961E-02	1.00090942E-03	3.59369540	6.23078639E-02	7.11420068E-02	2.64415511E-03	17.30529860	0.583391864	4.26109294E-16	152.71 ± 7.21
29	1.17760318E-01	3.35536012E-03	8.62640880	1.99948399E-01	1.58036764E-01	6.44908067E-03	21.89308040	1.313797540	2.52994757E-16	191.12 ± 12.68
29	1.25442097E-01	2.76290239E-03	2.09988952	4.60509217E-02	1.01953022E-01	5.59116105E-03	21.54869730	0.829046980	1.54381717E-16	188.26 ± 9.41
29	3.72919893E-02	5.34898404E-04	4.01751713	5.00156297E-02	6.60600813E-02	1.12890874E-03	15.36294280	0.373372154	4.39879147E-16	136.2 ± 5.69
29	4.78879447E-02	8.05579259E-04	1.51212697	2.33229423E-02	4.96087955E-02	1.64988562E-03	19.03971630	0.664924003	5.94057717E-16	167.32 ± 8.01
29	2.96413087E-02	7.37221577E-04	4.12191169	5.00999971E-02	4.93279202E-02	1.24156025E-03	21.41894540	0.645906871	1.28169970E-15	187.18 ± 8.34
29	2.64825279E-02	3.73855330E-04	4.79232680	5.61432387E-02	4.94401807E-02	1.41697650E-03	21.05462210	0.603590573	1.16480696E-15	184.16 ± 8.05
29	7.75514123E-02	1.48167438E-03	1.16927068	1.97826913E-02	8.63377411E-02	2.71440973E-03	22.47896670	0.790980982	2.73193963E-16	195.96 ± 9.34
29	4.85876190E-02	6.28807397E-04	3.38459801	4.40699982E-02	5.52082638E-02	1.15551504E-03	18.37164420	0.588845076	6.09134370E-16	161.71 ± 7.45
29	6.38421325E-02	1.20026425E-03	3.16709177	4.39488338E-02	6.71997323E-02	2.11871561E-03	18.92144780	0.681644420	2.73581812E-16	166.33 ± 8.08

Read 7.35626170E-02 as 0.073562610.

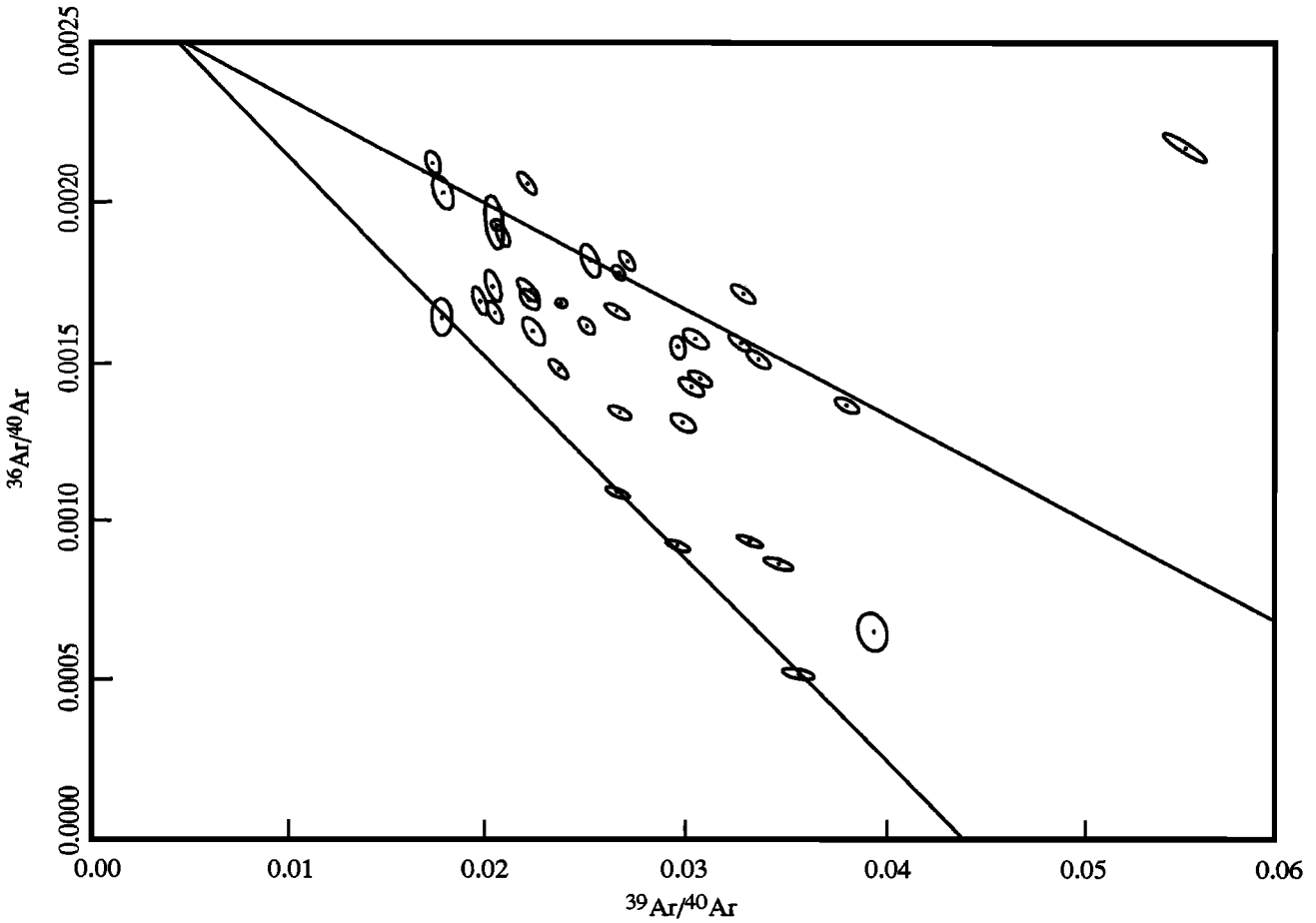


Fig. 11. $^{39}Ar/^{40}Ar$ versus $^{36}Ar/^{40}Ar$ isotope correlation diagram with 37 laser probe analyses from the Hound Island Volcanics. The field of most data is bound by two chords. The lower chord is defined by four points and has a $^{40}Ar/^{39}Ar$ ratio of 22.6 ± 0.2, which corresponds to an age of 197 ± 7 Ma, and an $^{40}Ar/^{36}Ar$ ratio of 363.7 ± 6.0. The upper chord is defined by 13 points, with a $^{40}Ar/^{39}Ar$ ratio of 12.1 ± 0.4, corresponding to an age of 108 ± 11 Ma, and an $^{40}Ar/^{36}Ar$ ratio of 381.5 ± 6.6.

The $^{37}Ar_{Ca}/^{39}Ar_K$ and $^{38}Ar_{Cl}/^{39}Ar_K$ values range from 1.0 to 9.1 and 0.03 to 0.15, corresponding to Ca/K and Cl/K values of 1.8 to 16.4 and 0.01 to 0.03, respectively. Contours of the isotope correlation diagram in terms of the $^{37}Ar_{Ca}/^{39}Ar_K$ and $^{38}Ar_{Cl}/^{39}Ar_K$ values of each analysis (Figures 12*a* and 12*b*, respectively) suggest that the older radiogenic-rich (low $^{36}Ar/^{40}Ar$) region of the triangle yields the highest Ca/K and the lowest Cl/K values and that the younger more atmospheric-enriched portion of the triangle yields lower Ca/K and higher Cl/K values. To determine whether or not this correlation is significant, we calculated the Ca/K values from electron microprobe analyses of the various mineral phases present in these samples.

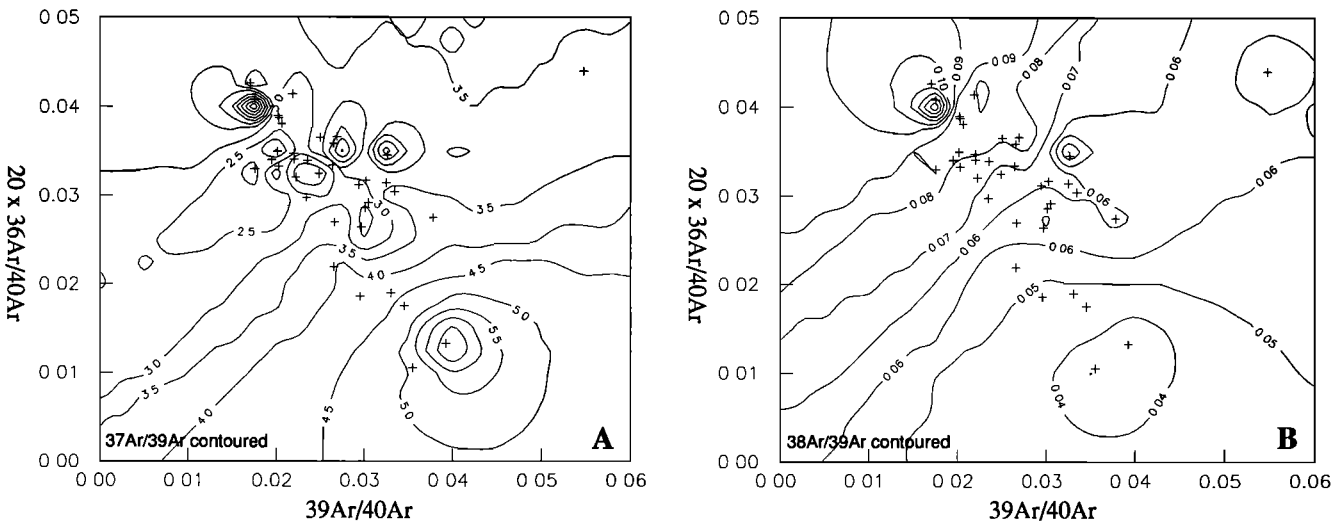


Fig. 12. $^{39}Ar/^{40}Ar$ versus $^{36}Ar/^{40}Ar$ isotope correlation diagram with contoured (*a*) $^{37}Ar/^{39}Ar$, which is proportional to the Ca/K ratio, and (*b*) $^{38}Ar/^{39}Ar$, which is proportional to the Cl/K ratio. Data points are denoted with a plus symbol. Note that the value on the $^{36}Ar/^{40}Ar$ axis is 20 times its actual value.

We examined replicate samples with backscattered electron images from a scanning electron microscope (SEM) and obtained standardless semiquantitative chemical analyses with an energy dispersive analysis system (EDA). Thin sections were carbon coated and examined with an International Scientific Instruments WB-6 operating at 15 kV; the EDA were collected with a Princeton Gamma Tech System 4. The EDA analyses of unaltered pyroxene indicate a high calcium content (an average of 1.0 when normalized to six oxygens for 11 analyses of two samples) and no detectable potassium. Thus the Ca/K ratio for the clinopyroxene is infinite.

The three distinct metamorphic alteration products of the clinopyroxenes are sphene, ferric biotite, and a Ca-poor, Fe, Mg aluminosilicate (cummingtonite?). EDA analyses of the sphene yielded 11-12 wt % calcium and no potassium, whereas analyses of the cummingtonite(?) and ferric biotite indicated <1 wt % and 5-10 wt % potassium, respectively (five analyses of one sample). The Ca/K for the sphene is infinite, whereas that of the ferric biotite is less than one.

Backscattered electron images of the feldspars exhibit intense alteration, and EDA analyses suggest that they are albitized in some locations. Highly birefringent inclusions have a high-potassium aluminosilicate composition similar to sericite, although small abundances of some atomic species precluded a pure analysis. Other Fe, Mg aluminosilicates and sphene occur as alteration products of what had been primary inclusion phases. The Ca/K ratios obtained from 30 EDA analyses of two samples range from 1 to 100 with an average of 20, whereas that of the sericite is much less than one.

There was no detectable chlorine in the EDA analyses; therefore any conclusions concerning the origin of the ^{38}Ar is purely speculative. On the basis of experimental studies [*Volfinger et al.*, 1985], however, we suspect that the chlorine would be concentrated in the Fe-rich hydrous phases (i.e., the ferric biotite and cummingtonite(?)).

These same hydrous phases are calcium poor and will tend to contain greater atmospheric contamination than the anhydrous clinopyroxene, sphene, and plagioclase. Therefore we interpret the high Cl/K, low Ca/K, and high atmospheric portion of the diagram as reflecting degassing primarily of the pyroxene alteration phases. The low Cl/K, high Ca/K, radiogenic-rich portion of the diagram then corresponds to degassing of the altered feldspar/sericite phases. Because the Ca/K contours cross the linear mixing lines between radiogenic and atmospheric argon, we have not interpreted any of the linear trends as isochrons but believe they represent mixing lines between mineral phases with varying age and atmospheric composition.

We therefore consider the lower chord of the triangle on Figure 11 as a minimum age for the main metamorphic crystallization event affecting these samples. This age of 197 ± 7 Ma is slightly younger than the Carnian-Norian age of the Hound Island Volcanics determined from age diagnostic macrofossils. The Rhaetian is now considered to be Norian by some North American paleontologists [*Tozer*, 1979], and therefore the Carnian-Norian paleontologic age corresponds to absolute ages of 230-208 Ma [*Kent and Gradstein*, 1984]. For this reason, we suggest that the main metamorphism of the Hound Island Volcanics occurred by hydrothermal alteration on the seafloor, not by metamorphism during the Late Cretaceous deformation. The work of *Samson et al.* [1989] indicates a mantle separation (Nd/Sm) age of 220 Ma for the Hound Island Volcanics, but the analytical error is a few tens of millions of years.

We further suggest that the upper chord on Figure 11 is a maximum age (108 ± 11 Ma) of a younger overprint that partially reset the argon system of the alteration phases by diffusion or recrystallization or both. We think this maximum age of the younger thermal event probably reflects overprinting of the Hound Island Volcanics associated with the deformation that occurred roughly between 100 and 90 Ma. Our finding of only normal polarity directions is consistent with this age of the overprint, because it would have been acquired during the Cretaceous normal polarity superchron.

The peak temperature during the early Late Cretaceous overprinting episode may be estimated assuming that the sericite and biotite define the field of data between the two chords in the correlation diagram. If the duration of the episode was approximately 20 m.y. and the sericite lost only 20% of its radiogenic argon by volume diffusion during that episode, then utilizing the diffusion parameters of *Robbins* [1972] from muscovite (40 kcal/mol, 1.38×10^{-4} cm^2/s assuming a cylindrical geometry) and 20 μm radius of diffusion yields a peak temperature of about 250°C [*Dodson*, 1975]. If the iron-rich biotite lost 90% of its radiogenic argon during the same thermal event, then the argon diffusion parameters of *Harrison et al.* [1985] (47 kcal/mol, $7.7 \times 10^{-2} cm^2/s$ assuming a cylindrical geometry) and a 10 μm radius of diffusion would suggest that the peak temperature was 270°C. Given the uncertainties in the parameters and the assumptions made, we consider these two estimates indistinguishable and suggest that the peak temperature of the reheating episode was around 250°-300°C.

Relationship of the New Results to the Previous Study

As discussed in the introduction, the previous study of *Hillhouse and Grommé* [1980] appeared reliable except for its puzzling tectonic implications. After removal of a low-stability component attributable to the PDF, their data passed fold and reversal tests. However, of the 12 sites they retained for their final calculations, three sites had in situ directions similar to our component B and nine sites had directions similar to our component C (see Figure 13). Moreover, the reversed polarity (component B) sites were from one area (Hound Island) and the normal polarity (component C) sites were from another area (Hamilton Island) (Figure 13). Thus we think that the stable directions they obtained in their study were 23 Ma (component B) and Cretaceous (component C) overprints that fortuitously passed fold and reversal tests.

Hillhouse and Grommé [1980] were unlucky in several respects. Had they collected samples from Turnabout Island and Kupreanof Island at Cape Bendel they probably would have found component A, which we interpret as primary. Had they collected more sites on Hound Island, they probably would have found component C there as well, which would have produced a negative fold test. Finally, if they had used predominantly thermal demagnetization instead of AF demagnetization, which we found does not isolate the characteristic remanence as well as thermal, their data might also have failed the fold test. We found significantly different best fit directions for three of their four sites from Hound Island. The difference in directions is particularly noticable when comparing the orientation of reversed-polarity sites from Hound Island and Kuiu Island in Figure 13. When we combine our reversed directions from Hound Island with those of *Hillhouse and Grommé* [1980] from Hamilton Island, the data fail the fold test.

It is easy to perceive these things in hindsight. However, given the same set of data as the reconnaissance study of *Hillhouse and Grommé* [1980], we probably would have come to the same conclusions. The keys to solving this paradox of a good paleomagnetic study and a troubling tectonic interpretation were wider sampling, greater use of thermal demagnetization, and $^{40}Ar/^{39}Ar$ geochronology.

TECTONIC IMPLICATIONS

Hound Island Volcanics Primary Direction

The primary direction in the Hound Island Volcanics indicates that the Alexander terrane was at a paleolatitude of 19.2° ± 10.3° in Late Triassic time. (When determining the statistical error on the paleolatitude, displacement and rotation we used the method of *Demarest* [1983] with a constant factor C = 0.78). This paleolatitude is in the same range of Late Triassic paleolatitudes (8°-19°) determined for all other portions of Wrangellia (Table 1) and is consistent with a Pennsylvanian geologic tie between the two terranes claimed by *Gardner et al.* [1988].

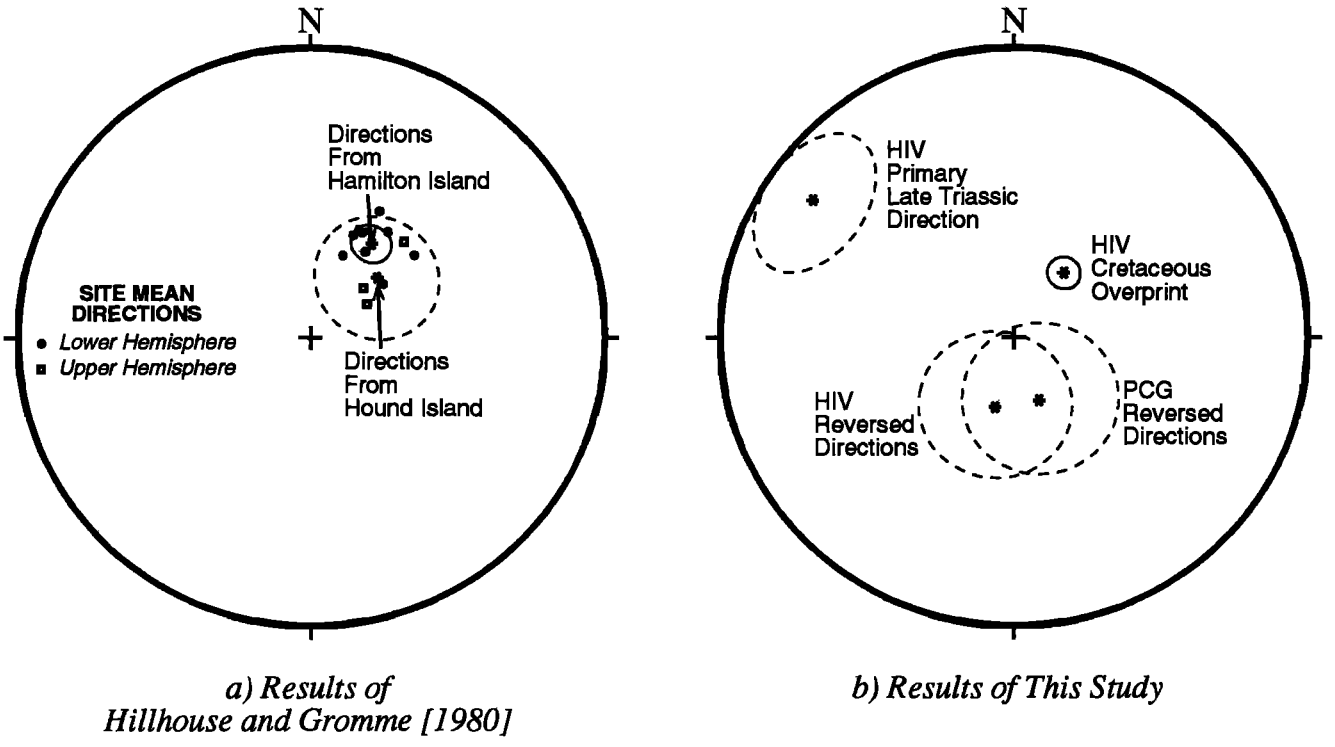


Fig. 13. Equal-area stereonets of *in situ* paleomagnetic results from (*a*) *Hillhouse and Gromme* [1980] and (*b*) this study. If the mean direction lies in the lower hemisphere, the α_{95} circle of confidence is shown as solid; if it lies in the upper hemisphere, it is dashed.

Using the Late Triassic reference pole closest in age to the Hound Island Volcanics (the combined Manicouagan-Chinle pole latitude = 58.4°N, longitude = 84.4°E, A_{95} = 7.0°) for North America favored by *May and Butler* [1986] (see Figure 14), we calculate the northward displacement of the Alexander terrane as 10.8° ± 11.6° and the rotation as 138.0° ± 12.6° clockwise relative to North America, if the terrane lay in the northern hemisphere in Late Triassic time. If it lay in the southern hemisphere, the relative northward displacement is 48.5° ± 11.6° and the rotation is 42.0° ± 12.6° counterclockwise. The ambiguity arises because the polarity of the Earth's magnetic field during eruption of the flows is not known and the terrane may have lain at either 19.2°N or 19.2°S in Late Triassic time.

The Alexander terrane lies outboard of the Stikine and the Cache Creek terranes. The Cache Creek terrane consists of strata with "oceanic" affinities (pillow basalts and radiolarian cherts) which indicates there was at least one ocean basin lying between the Alexander terrane and North America. In addition, the deep-marine strata of the Gravina-Nutzotin belt indicate an ocean basin between the Alexander and Stikine terranes. Closing of these basins may have entailed large displacements. If we assume that the Alexander terrane was in the northern hemisphere in Late Triassic time and moved northward only 11°, the amount of rotation indicated by our paleomagnetic data is very large, 138° clockwise. There is no evidence of intraterrane strike-slip faults of the proper age, size, and orientation that might allow vertical axis block rotations of 138°, and thus, if such a large rotation did occur, it must have been primarily due to transport on an oceanic plate.

Alternatively, the Alexander and Wrangellia terranes may have been in the southern hemisphere [*Panuska and Stone*, 1985]. Not only our paleomagnetic data for the Alexander terrane, but all the paleomagnetic data for Wrangellia suggest very large rotations for the northern hemisphere option (Table 1). The only exception is the "Seven Devils" terrane in southeastern Oregon [*Jones et al.*, 1977]; however, following *Sarewitz* [1983] we do not consider this a portion of Wrangellia. Therefore these rotations are greatly reduced if the terranes were in the southern hemisphere in the Late Triassic. For example, the Vancouver Island portion of Wrangellia would require 158° of counterclockwise rotation and almost no displacement if it was in the northern hemisphere in the Late Triassic. However, if Wrangellia was in the southern hemisphere the amount of rotation would be reduced to 22° clockwise.

Paleobiogeographic indicators also suggest that the Alexander terrane was in the southern hemisphere in Late Triassic time. An unusual species of the Norian bivalve of genus *Septocardia* was found in Alexander terrane rocks in the Keku Strait [*Newton*, 1983]. Although *Septocardia* are common in the North American cordillera, the only other known occurrence of the Alexander terrane species is in Peru [*Newton*, 1983]. If the locality of the *Septocardia* in Peru is not displaced with respect to South America, it would have been at a latitude of 15° south in Late Triassic time, similar to the paleolatitudes of Wrangellia and the Alexander terrane. Also, the distribution of the Late Triassic bivalve of genus *Monotis* suggests an affinity of the Alexander terrane with South America [*Silberling*, 1985].

Therefore the paleontological evidence and the total amount of rotation implied by the paleomagnetic data might favor a southern hemisphere position of the Wrangellia and Alexander terranes in Late Triassic time. There are some difficulties in proposing that the Alexander and Wrangellia terranes were in the southern hemisphere when we also consider mid-Cretaceous paleomagnetic data from the Keku Strait and elsewhere in the western Canadian Cordillera; we discuss these problems later.

Hound Island Volcanics and Pybus Formation Late Cretaceous Overprint Direction

We interpret component C (NE and down) in the Hound Island Volcanics as an early Late Cretaceous overprint as discussed above. We also interpret the characteristic direction of the Permian Pybus Formation

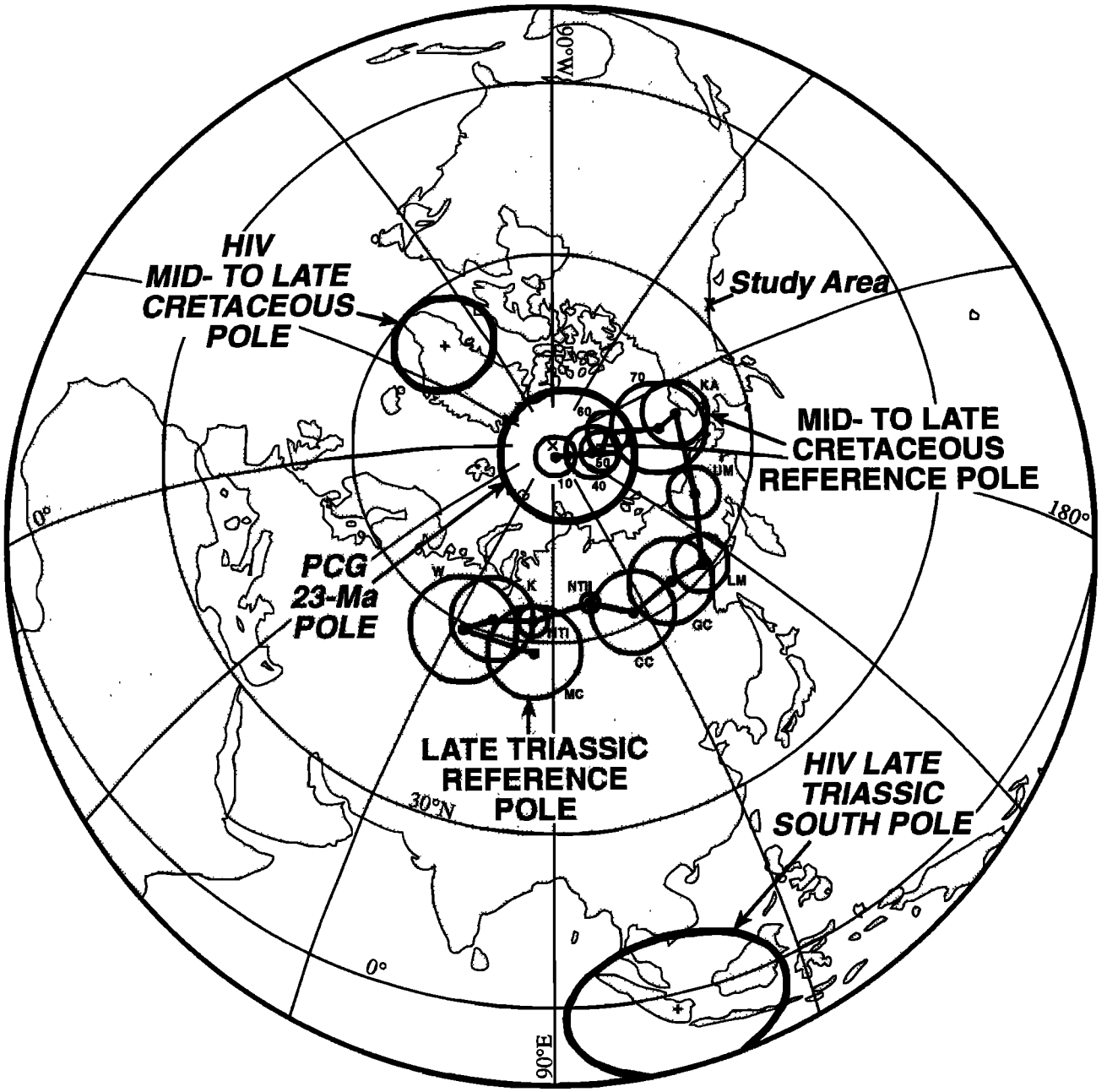


Fig. 14. Plot of the North American apparent polar wander path (APWP) and the paleomagnetic poles from this study and *Haeussler et al.* [this issue]. Abbreviations for reference poles are HIV, Hound Island Volcanics; PCG Point Camden gabbro (see *Haeussler et al.* [this issue]); MC, Manicouagan and Chinle (220 Ma); W, Wingate (200-206 Ma); K, Kayenta (194-200 Ma); NTI, Newark Trend Group I (195 ± 4 Ma); NTII, Newark Trend Group II (179 ± 3 Ma); CC, Coral Canyon (172 ± 6 Ma); GC, Glance Conglomerate (151 ± 2); LM, Lower Morrison (~149 Ma); UM, Upper Morrison (~145 Ma); KA, Cretaceous average (129-85 Ma); 70, 70 Ma; 60, 60 Ma; 50, 50 Ma; 40, 40 Ma; 10, 10 Ma. The APWP, and ages listed after the pole abbreviations, uses *May and Butler*'s [1986] poles for the Mesozoic, with *Globerman and Irving*'s [1988] Cretaceous average pole, and *Coe et al.*'s [1985] summary of the Tertiary APWP. In order to display the reference APWP and the poles from this study and *Haeussler et al.* [this issue], the Hound Island Volcanics Late Triassic *south* paleomagnetic pole is shown.

dolomite as an early Late Cretaceous overprint, because the mean direction is very similar to component C in the Hound Island Volcanics (not significantly different at 95% confidence). We do not think this is an overprint caused by intrusion of the 23-Ma Point Camden gabbro because the outcrops we sampled are not near any of the large intrusions, and because the mean direction is different at 95% confidence from the Point Camden gabbro mean direction.

Correct tectonic interpretation of any overprint direction is contingent upon knowing paleohorizontal at the time the overprint was acquired. Because paleohorizontal is difficult to ascertain, many studies assume the present horizontal is the same as the paleohorizontal. If we make this assumption, our data from the early Late Cretaceous overprint in the Hound Island Volcanics and the Pybus Formation imply large-scale northward displacement and clockwise rotation (a northward displacement of 17.3° ± 6.7° and a rotation of 63.2° ± 14.6 clockwise, using the reference pole of *Globerman and Irving* [1988]; see Figure 14). The amount of displacement and rotation is insensitive to uncertainties in the age of the overprint magnetization, because there was very little appar-

ent polar wander during the Cretaceous normal polarity superchron; the same reference pole is used from approximately 130 to 85 Ma.

These values of rotation and displacement are similar to, but slightly less than, those determined for 110-100 Ma plutons in the Coast Plutonic Complex [*Irving et al.*, 1985]. The pluton data imply that Baja British Columbia was at the latitude of Baja California, and has since undergone a northward displacement of 21° ± 6° with a clockwise rotation of 67° ± 8°. The amount of displacement and the direction and amount of rotation we determine from the Hound Island Volcanics overprint are within the error of that for the Coast Plutonic Complex data. In addition, the amount of displacement and the direction and amount of rotation inferred from the Hound Island Volcanics overprint are very similar to that indicated by 104-Ma bedded volcanic rocks of the Spences Bridge Group (15.5° ± 7.1° northward displacement, 66.3° ± 11.8° clockwise rotation), which lie inboard on the Intermontane terrane, about 1000 km to the southeast, in British Columbia [*Irving and Thorkelson*, 1990]. The data from the Spences Bridge Group do not suffer from problems with paleohorizontal, and they pass standard stability tests and appear reliable, although there were some complications in isolating the primary remanence direction.

Thus, assuming postmagnetization tilting is negligible, our results support and extend 350 km to the northwest the conclusions of *Irving et al.* [1985], which indicate the Coast Plutonic Complex and the Alexander and Stikine terranes (which are attached to all the terranes in the Intermontane and Insular belts) were at a lower latitude in mid-Cretaceous time. *Irving et al.* [1985] coined the term "Baja British Columbia" to describe this superterrane ("Baja" is the Spanish language equivalent of lower). They argued that paleohorizontal is the same as the present horizontal in the plutons on the basis of a consistency argument: paleomagnetic results along an 800 km distance of the Coast Plutonic Complex and into the North Cascades all indicate about the same amount of displacement. *Irving et al.* [1985] found it hard to imagine a mechanism that would tilt such a large area the same amount and in the same direction.

Butler et al. [1989] have resurrected the opposing view and contend these paleomagnetic directions reflect regional tilting of large portions of the Coast Plutonic Complex. They propose no specific mechanism but instead support their argument chiefly by citing geologic studies that show higher metamorphic grades east of the plutons that were studied paleomagnetically, a pattern which could be caused by east-side-up tilting. Note, however, that because the eastern ends of these plutons lie nearer the center of the Coast Plutonic Complex, an alternative explanation may be that there were higher temperatures [*Wood et al.*, 1991], and thus higher grade metamorphic rocks, toward the center of the Coast Plutonic Complex. Finally, *Butler et al.* [1989] argue that the age of paleomagnetic overprints used by *Irving et al.* [1985] to support large northward displacements are uncertain and should not be assumed to be about 100 Ma.

Is the Early Late Cretaceous Overprint Direction in the Hound Island Volcanics and Pybus Formation Tilted?

In view of contentions [*Butler et al.*, 1989] that plutons elsewhere in western Canada have been tilted, we examine more closely data relevant to the tilting hypothesis. A regional tilt-axis azimuth of roughly 330° is the most likely if the tilting mechanism was either isostatic uplift of the Coast Ranges during the Tertiary or contraction during the Late Cretaceous to early Tertiary. The average orientation of early Late Cretaceous fold axes in this portion of southeastern Alaska is roughly 330° [*Haeussler*, 1991], and the trend of the Coast Mountains in southeastern Alaska is also about 330°.

The structural geology of the Keku Strait argues against a regional northeast-side-up tilt. *Loney* [1964] described the large-scale structure as a "synclinorium" that has an axis that trends north-northwest—southeast. The broad synclinal structure is indicated by younger rocks toward the center of the Keku Strait as well as by the orientation of bedding in various units, including the Hound Island Volcanics. The rocks in the Keku Strait do not young to the southwest, which would be expected if there had been regional tilting.

Geobarometric and thermochronometric data have been used to argue for postfolding tilting in many areas; however, no such data exist for the region around the Keku Strait. Metamorphic grade information can be used to check the tilting hypothesis. *Haeussler* [1991] found that the rocks between the Keku Strait and about 30 km to the northeast were all metamorphosed to prehnite-pumpellyite facies, with the exception of Devonian(?) greenschist to amphibolite facies rocks that were probably metamorphosed prior to Permian time [*Muffler*, 1967]. Farther to the northeast, the rocks are upper greenschist to lower amphibolite facies; but these higher-grade rocks appear to be juxtaposed with the lower grade rocks along a fault with northeast-side-up movement. The abruptness of the transition between metamorphic facies does not appear to reflect simple tilting of a large block. Rocks to the southwest of the Keku Strait were probably metamorphosed to no higher than prehnite-pumpellyite facies [*Dusel-Bacon et al.*, 1991]. The lack of a gradual northeastward increase in the metamorphic grade around the Keku Strait argues against any significant regional tilt.

We are also unaware of structures in the area that could have accommodated significant tilting. Some models for regional tilting postulate "panels" that tilt individually, like dominos. Under this hypothesis we would expect that the trend of the faults between the panels would be parallel to the trend of the tilt axis, the motion on the faults would be dip slip with normal offset, and the fault plane would not be vertical. There are many faults in southeastern Alaska that trend parallel to the projected tilt axis. However, none of these faults have a significant component of dip slip movement; they are dextral-slip faults [*Haeussler*, 1991]. In addition, these faults have a subvertical dip, which is inconsistent with a dip-slip fault that was rotated to a shallower angle during tilting.

Despite the lack of geologic evidence for regional tilting, if we assume the overprint direction in the Hound Island Volcanics was deflected due to regional tilting, we can examine possible geometries of tilting with our paleomagnetic results. If tilting occurred in this region, the Point Camden gabbro provides an important constraint, because it intrudes the Hound Island Volcanics and has the expected direction for North America at 23 Ma [*Haeussler et al.*, this issue] (Figure 14). Therefore any significant tilting in the Keku Strait must have occurred prior to 23 Ma.

If we assume the overprint direction was deflected from the expected direction simply by regional tilting, the tilt axis azimuth would be 339° and the amount of tilt would be 20° northeast side up. The trend of this axis is similar to the expected tilt axis.

Another possibility includes orientation data from the Tertiary (Paleocene?) Kootznahoo Formation, which dips 13° to the south (strike of 087°) in the Keku Strait. The dip of the Kootznahoo Formation could mean that a local tilt correction should be applied to the overprint direction in the Hound Island Volcanics. The "locally-tilt-corrected" overprint direction is declination (Dec) = 69.7°, inclination (Inc) = 74.1°. This direction implies almost no northward displacement (5.7° ± 6.7°) and a significant clockwise rotation (54.1° ± 16.5°). In order to attribute this tilt-corrected overprint direction to regional tilting, the tilt would be 20° east-side-up about an axis trending N14°E, which is 44° clockwise from the expected trend. Therefore we consider this possibility unlikely. Alternatively, the dip of the Kootznahoo Formation might be primary. *Loney* [1964] describes the Kootznahoo Formation as an alluvial fan deposit with medium to very coarse grained sandstones and common conglomerate clasts and lenses. *Loney* [1964, p. 93] envisions the deposition of the Kootznahoo Formation as follows: "The rugged mountains cut in the folded Mesozoic and Paleozoic rocks adjacent to this basin poured

down coarse detritus into it, probably forming steep alluvial fans through which cut steep gradient streams." An initial dip of 13° is on the high side for an alluvial fan [*Kochel and Johnson*, 1984] but is not unheard of. We consider it likely that much of the 13° is initial dip, because the strike of the Kootznahoo Formation deviates 60° from the regional structural grain.

In addition to regional and local tilting of the overprint direction, two other mechanisms that could cause a difference between the the observed and expected directions are vertical axis rotations and poleward displacement. We have examined all plausible combinations of these four mechanisms and, not surprisingly, find that there is insufficient information in the paleomagnetic data alone to distinguish between the different possibilities. Therefore one needs to introduce other information and personal preferences. We suspect that the dip of the Kootznahoo Formation is mostly primary for the reasons discussed above, and that 12° to 16° of northward displacement occurred since mid-Cretaceous time. This second assertion is based upon the recent paleomagnetic studies of *Irving and Thorkelson* [1990] and *Marquis and Globerman* [1988] on 104-Ma and 70-Ma bedded volcanic rocks of the Spences Bridge Group and the Carmacks Volcanics, respectively, both from the Intermontane superterrane. These studies indicate 15.5° ± 7.1° and 11.5° ± 8.1° of northward displacement, respectively (we have recalculated the displacement of *Marquis and Globerman* [1988] using the reference pole of *Gunderson and Sheriff* [1991]). Because the sampling localities lie on the Intermontane superterrane, which lies inboard of the Insular superterrane, the paleomagnetically inferred displacements should apply to both superterranes. The paleomagnetic data from these studies pass standard stability tests and are not easily discounted, despite some concern about the redundancy in sampling similar directions in the Earth's secular variation in the Carmacks Volcanics [*Butler*, 1990]. In addition, the amount of displacement indicated by the Hound Island Volcanics and Pybus Formation in situ direction (17.3° ± 6.7° northward) is within the confidence limits of the displacements implied by both the Spences Bridge Group and the Carmacks Volcanics. As discussed above, there is no convincing geologic evidence for regional tilting in the vicinity of the Keku Strait, and thus we do not easily accept the hypothesis that the Cretaceous overprint in the Hound Island Volcanics and the Pybus Formation was tilted. Therefore we favor a scenario in which there was 12° to 16° of northward displacement between 100 and ~50 Ma, followed by some tilting of the region.

We do not entirely discount regional tilting during intrusion and rapid uplift of the Coast Mountains Plutonic Complex [*Hollister*, 1982], because deeply buried rocks in the crystalline core of the Coast Plutonic Complex presently crop out at the surface. Although these rocks could have been brought to the surface entirely along faults such as the Coast Ranges Megalineament [*Brew and Ford*, 1977], this seems improbable. Thus some regional tilting appears likely, but because the Keku Strait is about 60 km to the southwest of the Coast Plutonic Complex, the effect probably is rather small. Further evaluation of this hypothesis awaits collaborative studies between paleomagnetists, metamorphic geologists, structural geologists, and $^{40}Ar/^{39}Ar$ geochronologists.

Permian Pybus Formation Dolomite

A northeast and down direction was ubiquitous on Zijderveld vector diagrams from samples of the Permian Pybus Formation, although many samples had demagnetization paths that were heading toward a direction that is much shallower or of reversed polarity (see Figure 9). (However, note that because the overprint direction is steep, most other directions are likely to be shallower.) In order to better isolate this unremoved component of magnetization, we determined great circles for the demagnetization path using the principal component method of *Kirschvink* [1980] (Figure 15). We also noted which direction along the great-circle path that the directions were headed, allowing us to determine which of the two poles to the best fit plane through the poles was the direction of the unresolved component.

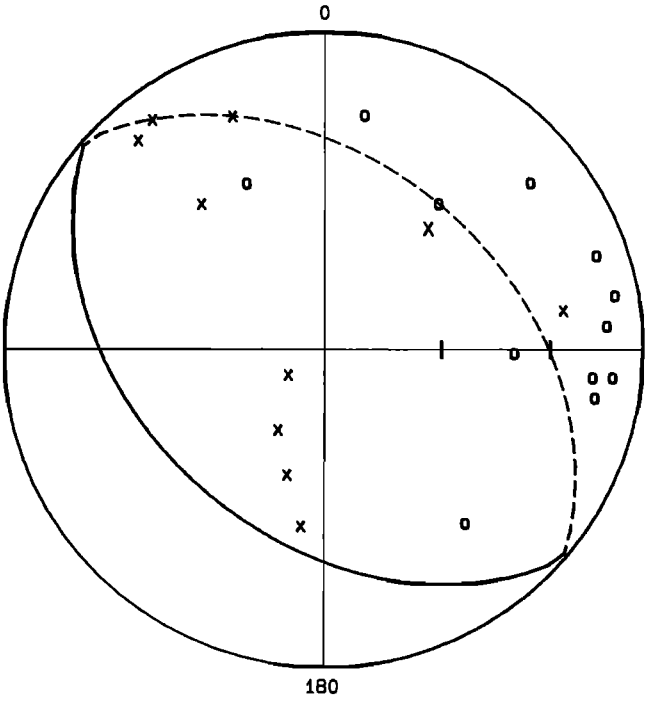


Fig. 15. Poles to best fit great circles determined from demagnetization paths of samples in the Pybus Formation. Upper and lower hemisphere directions are shown with circles and crosses, respectively. The best fit great circle through these poles gives the orientation of the unisolated component. The great circle is not well constrained (maximum angular deviation, MAD = 40.3), but gives an upper-hemisphere direction of Dec = 221.4°, Inc = -48.8° using sector constraints [*McFadden and McElhinny*, 1988].

The great-circle method of *McFadden and McElhinny* [1988] gives a direction of Dec = 221.4°, Inc = -48.8° with a maximum angular deviation (MAD) of 40.3°. Because the MAD is so large, we do not consider the actual intersection of the great circles as a reliable direction for the unresolved component, but the fact that the direction is of reversed polarity is probably significant. The age call for the Pybus Formation is Leonardian (Lower Permian) [*Brew et al.*, 1984], which is within the Permian Kiamen reversed polarity interval [*Harland et al.*, 1982]. Therefore, if the reversed direction indicated by the great circles is primary and the age of the rocks is correct, we can conclude that the Alexander terrane was in the northern hemisphere during Permian time. In addition, the Paleozoic fauna of the Alexander terrane have the closest affinities with North American and Siberian faunas [e.g., *Savage*, 1989], supporting a northern hemisphere position. A similar conclusion was reached by *Panuska and Stone* [1981] for Permian rocks from the Wrangellia terrane, which lies just to the northwest.

Gehrels and Saleeby [1987*b*] hypothesized that the Alexander terrane was associated with Australia during the Paleozoic because there is a striking similarity in the timing of orogenic events. If the Alexander terrane is constrained to the northern hemisphere in Permian time as suggested by our data, then it is impossible for the Alexander terrane to be associated with Australia during Permian time, as Australia was part of Gondwana and certainly was in the southern hemisphere. In addition, the Nd/Sm data of *Samson et al.* [1989] indicate the Alexander terrane formed from mantle material rather than from preexisting continental crust, throughout its entire Paleozoic and Mesozoic history. These data

are inconsistent with an Australian birthplace of the Alexander terrane, because the age of the basement rocks in eastern Australia is Proterozoic, and they found no Proterozoic Nd/Sm model ages from the Alexander terrane rocks.

Toward a Displacement History of the Alexander Terrane From Permian Time to the Present

With the paleomagnetic data we have discussed we attempt to outline a displacement history of the Alexander terrane from Permian time to the present. As just mentioned, our data suggest the Alexander terrane was in the northern hemisphere in Permian time.

The displacement history between the Late Triassic and Late Cretaceous depends upon whether the Alexander terrane was in the northern or southern hemisphere in Late Triassic time, and both possibilities have difficulties. If we assume the in situ direction of the early Late Cretaceous overprint in the Keku Strait gives approximately the correct paleolatitude and orientation, both options must conclude with 17° northward displacement and 63° clockwise rotation relative to North America between ~100 Ma and the present time. As discussed previously, a southern hemisphere Late Triassic location might be favored when the total amount of rotation and the paleobiogeographic data are considered; however, this hypothesis becomes less attractive when the rotation between Late Triassic and mid-Cretaceous time is evaluated. In this scenario, the Alexander terrane would have moved south to the southern hemisphere between Permian and Late Triassic time and then would have undergone 31.2° of northward displacement and 105.2° counterclockwise rotation between Late Triassic and Late Cretaceous time. This kinematic history is unlike recent terrane trajectory models, which predict a small (~20°) clockwise rotation as the Alexander terrane rides passively toward North America on the Farallon plate between 180 and 100 Ma [*Debiche et al.* 1987]. If the displacement history indicated by our paleomagnetic data is even approximately correct, a very large counterclockwise rotation would have had to occur between 220 and ~180 Ma, a time for which oceanic plate motions are completely unconstrained.

Alternatively, if the Alexander terrane was in the northern hemisphere, it underwent a total of 7.2° of northward displacement with 75.0° of clockwise rotation relative to North America between Late Triassic and Late Cretaceous time. Under this hypothesis, the terrane could have been moving southward (not into the southern hemisphere) on an unknown plate between Late Triassic and Early Jurassic time before it was transferred to, and began moving northeastward on, the Farallon plate [*Debiche et al.*, 1987]. Right-lateral shearing interpreted by *McClelland and Gehrels* [1990] to have occurred in Early to Middle Jurassic time along the (present) northeastern margin of the Alexander terrane might have resulted from the transferring process. On the basis of terrane trajectory models, if the Alexander terrane had been riding passively on the Farallon plate, a moderate (~30°) clockwise rotation would be expected for the interval between 180 and 100 Ma [*Debiche et al.*, 1987]. The additional 45° of clockwise rotation required by our paleomagnetic data could have easily occurred between 220 and 180 Ma, and perhaps as well in part during the early stages of collision with North America. This yoyo-like movement is reminiscent of that suggested by *Panuska and Stone* [1981] on the basis of other paleomagnetic data, in which Wrangellia and the Peninsular terrane were in the northern hemisphere from late Pennsylvanian to Late Triassic time and then moved southward into the southern hemisphere before moving northward into the northern hemisphere again in Middle to Late Jurassic time. Another attractive aspect of the northern hemisphere displacement history is that the sense of rotation remains the same before and after mid-Cretaceous time, and the amount of rotation is much less than with the southern hemisphere option. Although this scenario yields a better fit between our data and terrane trajectory models, we note that large and sometimes inconsistent rotations are implied by the paleomagnetic data available from other portions of Wrangellia (Table 1).

We also note that the three mid-Cretaceous (~100 Ma) paleomagnetic results from the Insular and Intermontane superterranes (the overprint in the Hound Island Volcanics and in the Permian Pybus Formation (this study); plutons in the Coast Plutonic Complex [*Irving et al.*, 1985]; and the Spences Bridge Group volcanic rocks [*Irving and Thorkelson*, 1990]) all indicate a clockwise rotation of about 65°, whereas the result from the 70 Ma Carmacks volcanics shows no significant rotation [*Marquis and Globerman*, 1988]. These data imply that 65° of clockwise rotation occurred in the interval between ~100 and 70 Ma. The terrane trajectory models of *Debiche et al.* [1987] indicate that roughly 30° of clockwise rotation can be attributed to northward transport along the North American continental margin. A potential problem with this interpretation is that the Carmacks Volcanics also show displacement but no rotation. About 15° of clockwise rotation would be expected simply from transport along the British Columbia portion of the North American continental margin. However, the Carmacks Volcanics lie within a 14° bend in the Tintina-Rocky Mountain Trench fault system and other regional structures. The result may have been that as the Carmacks Volcanics started to move around the bend in the fault, the clockwise rotation that had been acquired was nullified. The remaining ~35° of clockwise rotation not attributable to northward transport along the North American continental margin could have been caused by the collision of the stick-shaped Alexander terrane with North America. This "scissors style" of collision would be expected to have produced a south to north progression in the age of deformation. We know of no evidence for or against this hypothesis at the present time.

Finally, the Alexander terrane arrived in its present position with respect to North America around 50 Ma. A number of reliable paleomagnetic studies in western British Columbia and in southern Alaska indicate that the Intermontane and Insular superterranes were in place by about 50 Ma and have undergone essentially no rotation since that time [*Hicken and Irving*, 1977; *Fox and Beck*, 1985; *Bardoux and Irving*, 1989; *Symons and Wellings*, 1989; *Irving and Brandon*, 1990; *Vandall and Palmer*, 1990; *Hillhouse and Grommé*, 1982; *Hillhouse et al.*, 1985; *Coe et al.*, 1985; *Thrupp and Coe*, 1986; *Panuska et al.*, 1990].

CONCLUSIONS

We found three components of magnetization (A, B, and C) in the Hound Island Volcanics. Component A is the primary component that was acquired in Late Triassic time. A primary remanence is indicated chiefly by a positive fold test. We find this direction mostly in lava flows from one geographic area where the Hound Island Volcanics crop out. This component indicates a paleolatitude of 19.2° for the Alexander terrane, which is similar to those determined for portions of Wrangellia in Late Triassic time, suggesting that the two terranes were near to each other.

We interpret component B as an overprint caused by intrusion of the Point Camden Gabbro at 23 Ma. We find directions in the Hound Island Volcanics that are very similar to those in nearby outcrops of the Point Camden gabbro.

Component C is an early Late Cretaceous (between ~100 and 90 Ma) overprint caused by the mid-Cretaceous to early Tertiary orogenic event in southeastern Alaska. The date of this overprint is suggested by both geologic evidence and $^{40}Ar/^{39}Ar$ laser-probe ages from the Hound Island Volcanics. The Permian Pybus Formation has a similar overprint direction, which we consider to have the same age. If present horizontal is the same as the paleohorizontal at the time the overprint was acquired, the direction indicates a large northward displacement (17°) and clockwise rotation (63°) similar to those determined from 110- to 100-Ma

plutons in the Coast Plutonic Complex [*Irving et al.*, 1985], and from 104-Ma bedded volcanic rocks of the Spences Bridge Group. An alternative explanation for this direction is regional tilting about a horizontal axis, although there is no compelling geologic evidence for tilting of the study area.

Our interpretation of the previous study of the Hound Island Volcanics by *Hillhouse and Grommé* [1980] is that only components B and C were sampled, which fortuitously passed a fold and reversal test when the tilt correction was applied. We were able to resolve this problem through wider sampling, more extensive use of thermal demagnetization, and $^{40}Ar/^{39}Ar$ geochronology. Reflected light microscopy of the oxide minerals in the Hound Island Volcanics shows that the rocks with a primary magnetization were highly oxidized compared to those that were overprinted.

$^{40}Ar/^{39}Ar$ geochronology in conjunction with optical and electron probe studies indicate that the major metamorphism of the Hound Island Volcanics occurred as a result of hydrothermal alteration on the seafloor soon after deposition of the lavas.

Finally, great-circle analysis of directions in samples from the Permian Pybus Formation suggests that the Alexander terrane was in the northern hemisphere in Permian time.

Acknowledgments. Financial support for this research was provided by National Science Foundation grants EAR-8903898 and EAR-8417639, the Geological Society of America and the Earth Sciences Board at UCSC. This is Institute of Tectonics Paleomagnetism Laboratory contribution 143. We appreciate the encouragement of J. Hillhouse and S. Grommé to conduct this study. We gratefully acknowledge Kate Haeussler and Marc Crum who assisted in the field studies, and Kate Haeussler for tremendous moral support.

REFERENCES

Bardoux, M., and E. Irving, Paleomagnetism of Eocene rocks of the Kelwona and Castlegar areas, British Columbia: Studies in determining paleohorizontal; *Can. J. Earth Sci., 26*, 829-844, 1989.

Berg, H. C., D. L. Jones, and D. H. Richter, Gravina-Nutzotin belt—Tectonic significance of an upper Mesozoic sedimentary and volcanic sequence in southern and southeastern Alaska; *U.S. Geol. Surv. Prof. Pap. 800-D*, D1-D24, 1972.

Brew, D. A., and A. B. Ford, Megalineament in southeastern Alaska marks southwest edge of Coast Range batholithic complex; *Can. J. Earth Sci., 15*, 1763-1772, 1977.

Brew, D. A., and R. P. Morrell, Intrusive rocks and plutonic belts of southeastrn Alaska, U. S. A.; Circum-Pacific plutonic terranes, edited by J. A. Roddick, *Mem. Geol. Soc. Am. 159*, 171-193, 1983.

Brew, D. A., and L. P. J. Muffler, Upper Triassic undevitrified volcanic glass from Hound Island, Keku Strait, southeastern Alaska; *U.S. Geol. Surv. Prof. Pap. 525-C*, C38-C43, 1965.

Brew, D. A., T. A. Ovenshine, S. M. Karl, and S. J. Hunt, Preliminary reconnaissance geologic map of the Petersburg and parts of the Port Alexander and Sumdum 1:250,000 quadrangles, southeastern Alaska; *U.S. Geol. Surv. Open-File Rep. 84-405*, 1984.

Butler R. F., Northward motion of the Whitehorse Trough: Paleomagnetic evidence from the Upper Cretaceous Carmacks Group: Discussion; *Can. J. Earth Sci., 27*, 614-617, 1990.

Butler, R. F., G. E. Gehrels, W. C. McClelland, S. R. May, and D. Klepacki, Discordant paleomagnetic poles from the Canadian Coast Plutonic Complex, regional tilt rather than large-scale displacement?; *Geology, 17*, 691-694, 1989.

Coe, R. S., B. R. Globerman, P. W. Plumley, and G. Thrupp, Paleomagnetic results from Alaska and their tectonic implications: *Tectonostratigraphic Terranes of the Circum-Pacific Region, Earth Sci. Ser.*, vol. 1, edited by D. G. Howell, pp. 85-108, Circum-Pacific Council for Energy and Mineral Resources, Houston, Tex., 1985.

Cohen, H. A., T. C. Onstott, and N. Lundberg, and C. Hall, 40/39 Ar laser probe dating of detrital phenocrysts to constrain the age of volcanism, Gravina belt, SE Alaska (abstract); *Eos Trans. AGU, 71*, 1616, 1990.

Debiche, M. G., A. Cox, and D. C. Engebretson, The motion of allochthonous terranes across the North Pacific basin; *Spec. Pap. Geol. Soc. Am. 207*, 56 pp., 1987.

Demarest, H. H., Jr., Error analysis for the determination of tectonic rotation from paleomagnetic data; *J. Geophys. Res., 88*, 4321-4328, 1983.

Detterman, R. L., and B. L. Reed, Stratigraphy, structure, and economic geology of the Iliamna Quadrangle, Alaska; *U.S. Geol. Surv. Bull., 1368-B*, B1-B86, 1980.

Dodson, M. H., Kinetic processes and thermal histories of rocks; *Year Book Carnegie Inst. Washington, 74*, 210-217, 1975.

Dusel-Bacon, C., D. A. Brew, and S. L. Douglass, Metamorphic facies map of southeastern Alaska—Distribution, facies, and ages of regionally metamorphosed rocks; *U.S. Geol. Surv. Open-File Rep.: 91-29*, 1991.

Fischer, D. E., Rare gas abundances and MORB; *Geochim. Cosmochim. Acta, 51*, 2531-2541, 1986.

Fisher, Sir R., Dispersion on a sphere; *Proceedings of the Royal Society of London Ser. A., 217*, 295-305, 1953.

Fox, K. F., and M. E. Beck, Paleomagnetic results for Eocene volcanic rocks from northeastern Washington and the Tertiary tectonics of the Pacific Northwest; *Tectonics, 4*, 323-341, 1985.

Gardner, M. C., S. C. Bergman, G. W. Cushing, E. M. MacKevett, Jr., G. Plafker, R. B. Campbell, C. J. Dodds, W. C. McClelland, P. A. Mueller, Pennsylvanian pluton stitching of Wrangellia and the Alexander terrane, Wrangell Mountains, Alaska; *Geology, 16*, 967-971, 1988.

Gehrels, G. E., and J. B. Saleeby, Geology of southern Prince of Wales Island, southeastern Alaska; *Geol. Soc. Am. Bull., 98*, p.123-137, 1987a.

Gehrels, G. E., and J. B. Saleeby, Geologic framework, tectonic evolution, and displacement history of the Alexander terrane; *Tectonics, 6*, 151-173, 1987b.

Globerman, B. R, and E. Irving, Mid-Cretaceous paleomagnetic reference field for North America: Restudy of 100 Ma intrusive rocks from Arkansas; *J. Geophys. Res., 93*, 11,721-11,733, 1988.

Gunderson, J. A., and S. D. Sheriff, A new Late Cretaceous paleomagnetic pole from the Adel Mountains, west central Montana; *J. Geophys. Res., 96*, 317-326, 1991.

Haeussler, P. H., Paleomagnetic and structural constraints on the accretion history of the Alexander terrane, southeastern Alaska, Ph.D. dissertation, 213 pp., Univ. of Calif. Santa Cruz, 1991.

Haeussler, P. J., R. S. Coe, and P. Renne, Paleomagnetism and geochronology of 23-Ma gabbroic intrusions in the Keku Strait, Alaska, and implications for the Alexander terrane, *J. Geophys. Res.*, this issue.

Haggerty, S. E., Oxidation of opaque mineral oxides in basalts; *Oxide Minerals, Short Course Notes*, vol. 3, edited by D. Rumble III, pp. Hg1-Hg100, Mineralogical Society of America, Blacksburg, Virg., 1976.

Harland, W. B., A. V. Cox, P. G. Llewellyn, C. A. G. Pickton, A. G. Smith, and R. Walters, *A Geologic Time Scale*; 131 pp., Cambridge University Press, New York, 1982.

Harrison, T. M., I. Duncan, and I. McDougall, Diffusion of ^{40}Ar in biotite: Temperature, pressure, and compositional effects; *Geochim. Cosmochim. Acta, 49*, 2461-2468, 1985.

Hicken, A., and E. Irving, Tectonic rotation in western Canada; *Nature, 268*, 219-220, 1977.

Hillhouse, J. W., Paleomagnetism of the Triassic Nikolai Greenstone, McCarthy quadrangle, Alaska; *Can. J. Earth Sci., 14*, 2578-2592, 1977.

Hillhouse, J. W., and C. S. Grommé, Paleomagnetism of the Triassic Hound Island Volcanics, Alexander terrane, southeastern Alaska; *J. Geophys. Res., 85*, 2594-2602, 1980.

Hillhouse, J. W., and C. S. Grommé, Limits to northward drift of the Paleocene Cantwell Formation, central Alaska; *Geology, 10*, 552-556, 1982.

Hillhouse, J. W., and C. S.Grommé, Northward displacement and accretion of Wrangellia: New paleomagnetic evidence from Alaska; *J. Geophys. Res., 89*, 4461-4477, 1984.

Hillhouse, J. W., C. S. Grommé, and T. L. Vallier, Paleomagnetism and Mesozoic tectonics of the Seven Devils volcanic arc in northeastern Oregon; *J. Geophys. Res., 87*, 3777-3794, 1982.

Hillhouse, J. W., C. S. Grommé, and B. Csjety, Jr., Tectonic implications of paleomagnetic poles from Lower Tertiary volcanic rocks, south central Alaska; *J. Geophys. Res., 90*, 12,523-12,535, 1985.

Hollister, L. S., Metamorphic evidence for rapid (2 mm/yr) uplift of a portion of the Central Gneiss Complex, Coast Mountains, B. C.; *Can. Mineral., 20*, 319-332, 1982.

Irving, E., and M. T. Brandon, Paleomagnetism of the Flores volcanics, Vancouver Island, in place by Eocene time; *Can. J. Earth Sci., 27*, 811-817, 1990.

Irving, E., and D. J. Thorkelson, On determining paleohorizontal and latitudinal shifts: Paleomagnetism of Spences Bridge Group, British Columbia; *J. Geophys. Res., 95*, 19,213-19,234, 1990.

Irving, E., G. J. Woodsworth, P. J. Wynne, and A. Morrison, Paleomagnetic evidence for displacement from the south of the Coast Plutonic Complex, British Columbia; *Can. J. Earth Sci., 22*, 584-598, 1985.

Jones, D. L., and N. J. Silberling, Mesozoic stratigraphy—The key to tectonic analysis of southern and central Alaska; *U. S. Geological Survey Open-File Report 79-1200*, 37 pp., 1979.

Jones, D. L., N. J. Silberling, and J. Hillhouse, Wrangellia—A displaced terrane in northwestern North America; *Can. J. Earth Sci., 14*, 2565-2577, 1977.

Kent, D. V., and F. M. Gradstein, A Jurassic to Recent chronology; *The Geology of North America*, vol. M, *The Western Atlantic Region*, edited by B. E. Tucholke and P. R. Vogt, pp. 45-50, Geological Society of America, Boulder, Colo., 1984.

Kirschvink, J. L., The least-squares line and plane analysis of paleomagnetic data; *Geophys. J. R. Astron. Soc., 62*, 699-718, 1980.

Kochel, R. C., and R. A. Johnson, Geomorphology and sedimentology of humid-temperate alluvial fans, central Virginia; Gravels and Conglomerates, edited by E. Koster, and R. Steel, *Mem. Can. Soc.Pet. Geol. 10*, 109-122, 1984.

Lee, J. K. W., T. C. Onstott, and J. A. Hanes, An $^{40}Ar/^{39}Ar$ investigation of the contact effects of a dyke intrusion, Kapuskasing Structural Zone, Ontario; *Contrib. Mineral. Petrol., 105*, 87-105, 1990.

Loney, R. A., Stratigraphy and petrography of the Pybus-Gambier area, Admiralty Island, Alaska; *U.S. Geol. Surv. Bull. 1178*, 103 pp., 1964.

Marquis, G., and B. R. Globerman, Northward motion of the Whitehorse Trough, paleomagnetic evidence from the Upper Cretaceous Carmacks Group; *Can. J. Earth Sci., 25*, 2005-2016, 1988.

May, S. R., and R. F. Butler, North American Jurassic apparent polar wander: Implications for plate motion, paleogeography, and cordilleran tectonics; *J. Geophys. Res., 91*, 11,519-11,544, 1986.

McClelland W. C., and G. E. Gehrels, Geology of the Duncan Canal shear zone: Evidence for Early to Middle Jurassic deformation of the Alexander terrane, southeastern Alaska; *Geol. Soc. Am. Bull., 102*, 1378-1392, 1990.

McElhinny, M. W., Statistical significance of the fold test in palaeomagnetism; *Geophys. J. R. Astron. Soc., 8*, 338-340, 1964.

McFadden, P. L., A new fold test for paleomagnetic studies; *Geophys. J. Int., 103*, 163-169, 1990.

McFadden, P. L., and M. W. McElhinny, The combined analysis of remagnetization circles and direct observations in palaeomagnetism; *Earth Planet. Sci. Lett., 87*, 161-172, 1988.

McFadden, P. L., R. T. Merrill, M. W. McElhinny, and S. H. Lee, Reversals of the Earth's magnetic field and temporal variations of the dynamo families; *J. Geophys. Res., 96*, 3923-3933, 1991.

Monger, J. W. H., R. A. Price, and D. J. Tempelman-Kluit, Tectonic accretion of the two major metamorphic and plutonic welts of the Canadian Cordillera; *Geology, 10*, 70-75, 1982.

Muffler, L. J. P., Stratigraphy of the Keku Islets and neighboring pars of Kuiu and Kupreanof islands, southeastern Alaska; *U.S. Geol. Surv. Bull., 1241-C*, 52 pp., 1967.

Newton, C. R., Paleozoogeographic affinities of Norian bivalves from the Wrangellian, Peninsular, and Alexander Terranes, Western North America; *Pre-Jurassic Rocks in Western North American Suspect Terranes*, edited by C. H. Stevens, pp. 37-48, The Pacific Section Society of Economic Paleontologists and Mineralogists, Los Angeles, Calif., 1983.

Nokleberg, W. J., D. L. Jones, and N. J. Silberling, Origin and tectonic evolution of the Maclaren and Wrangellia terranes, eastern Alaska Range, Alaska; *Geol. Soc. Am. Bull., 96*, 1251-1270, 1985.

Onstott, T. C., and M. W. Peacock, Argon retentivity of hornblendes: A field experiment in a slowly cooled metamorphic terrane; *Geochim. Cosmochim. Acta, 51*, 2891-2903, 1987.

Panuska, B. C., and D. B. Stone, Late Paleozoic paleomagnetic data for Wrangellia: Resolution of the polarity ambiguity; *Nature, 293*, 561-563, 1981.

Panuska, B. C., and D. B. Stone, Latitudinal motion of the Wrangellian terranes and the Southern Alaska Superterrane, *Tectonostratigraphic Terranes of the Circum-Pacific Region, Earth Sci. Ser.*, vol. 1, edited by D. G. Howell, pp. 109-120, Circum-Pacific Council for Energy and Mineral Resources, Houston, Tex., 1985.

Panuska, B. C., D. B. Stone, and D. L. Turner, Paleomagnetism of Eocene volcanic rocks, Talkeetna Mountains, Alaska; *J. Geophys. Res., 95*, 6737-6750, 1990.

Robbins, G. A., Radiogenic argon diffusion in muscovite under hydrothermal conditions; Ms. thesis, 42 pp., Brown University, 1972.

Samson, S. D., and E. C. Alexander, Jr., Calibration of the interlaboratory $^{40}Ar/^{39}Ar$ dating standard, Mmhb-1; *Chem. Geol. (Isot. Geosci.), 66*, 27-34, 1987.

Samson, S. D., W. C. McClelland, P. J. Patchett, G. E. Gehrels, and R. G. Anderson, Evidence from neodymium isotopes for mantle contributions to Phanerozoic crustal genesis in the Canadian Cordillera; *Nature, 337*, 705-709, 1989.

Sarewitz, D., Seven Devils terrane: Is it really a piece of Wrangellia?; *Geology, 11*, 634-637, 1983.

Savage, N. M., Paleozoic biogeographic signals from the Alexander terrane; *Geol. Soc. Am. Abstr. Programs, 21*, 140, 1989.

Symons, D. T. A., and M. R. Wellings, Paleomagnetism of the Eocene Kamloops Group and the cratonization of Terrane I of the Canadian Cordillera; *Can. J. Earth Sci., 26*, 821-828, 1989.

Silberling, N. J., Biogeographic significance of the Upper Triassic bivalve Monotis in Circum-Pacific accreted terranes; *Tectonostratigraphic Terranes of the Circum-Pacific Region, Earth Sci. Ser.*, vol. 1, edited by D. G. Howell, pp. 63-70, Circum-Pacific Council for Energy and Mineral Resources, Houston, Tex., 1985.

Thrupp, G. A., and R. S. Coe, Early Tertiary paleomagnetic evidence and the displacement of southern Alaska; *Geology, 14*, 213-217, 1986.

Tozer, E. T., Latest Triassic ammonid faunas and biochronology, western Canada; *Geol. Soc. Can. Pap. 79-1B*, 127-135, 1979.

Turner, G., J. C. Huneke, F. A. Podosek, and G. J. Wasserburg, $^{40}Ar/^{39}Ar$ ages and cosmic ray exposure ages of Apollo 14 samples; *Earth Planet. Sci. Lett., 12*, 19-35, 1971.

Umhoefer, P. J., Northward translation of "Baja British Columbia" along the Late Cretaceous to Paleocene margin of western North America; *Tectonics, 6*, 377-394, 1987.

Vandall, T. A., and H. C. Palmer, Upper limit of docking time for Stikinia and Terrane I: Paleomagnetic evidence from the Eocene Ootsa Lake Group, British Columbia; *Can. J. Earth Sci., 27*, 212-218, 1990.

Van der Voo, R., Paleomagnetism of North America: The craton, its margins, and the Appalachian belt; Geophysical Framework of the Continental United States, edited by L. C. Pakiser and W. D. Mooney, *Mem. Geol. Soc. Am. 172*, 447-470, 1989.

Volfinger, M., J. L. Robert, D. Vielzeuf, and A. M. R. Neiva, Structural control of the chlorine content of OH-bearing silicates (micas and amphiboles); *Geochim. Cosmochim. Acta, 49*, 37-48, 1985.

Wood, D. J., H. H. Stowell, T. C. Onstott, and L. S. Hollister, $^{40}Ar/^{39}Ar$ constraints on the emplacement, uplift and cooling of the Coast Plutonic Complex sill, southeastern Alaska; *Geol. Soc. Amer. Bull., 103*, 9. 849-860, 1991.

Yole, R. W., and E. Irving, Displacement of Vancouver Island: Paleomagnetic evidence from the Karmutsen Formation; *Can. J. Earth Sci., 17*, 1210-1288, 1980.

Zijderveld, J. D. A., A. C. demagnetization of rocks: Analysis of results, *Methods in Paleomagnetism*; edited by D. W. Collinson et al., pp. 254-286, Elsevier, Amsterdam, 1967.

R. S. Coe, Earth Sciences Board and Institute of Tectonics, University of California, Santa Cruz, CA 95064.

P. J. Haeussler, U.S. Geological Survey, 4200 University Drive, Anchorage, AK 99508.

T. C. Onstott, Department of Geology and Geophysics, Princeton University, Princeton, NJ 08544.

(Received August 15, 1991;
revised March 28, 1992;
accepted June 2, 1992.)